

PACMINEX PTY. LIMITED

000732

MEMORANDUM TO: Mr. P.J. Curtis

OUR REF: DGT/SS

FROM: D.G. Tonkin

DATE: 27th November, 1974
Typed - 10/12/74

PANDURRA PETROGRAPHY

Could you please re-examine all the Pandurra sections; in the groups shown on the attached sheet.

Now that basic stratigraphy of the Pandurra area has been completed, the samples submitted previously can be assigned to their appropriate stratigraphic unit with more certainty.

You will notice that the number of specimens considered to be Whyalla Sandstone has been reduced from 16 to 6, for example.

What I would like is a description of the general characteristics of each of the rock units, noting especially any diagnostic features.

Similarities between groups should be pointed out.

Subdivision of groups would be appreciated.

Note : The attached list includes the latest batch of samples and a further batch of 3 in preparation now. The earlier samples were reported on by you in :-

1. Petrography of Pandurra Thin Sections I, PJC/IM dated 4/9/73.
2. PMR 5/74
3. PMR 73/74
4. PMR 87/74.

D.G. TONKIN

c.c. R.W. Gersteling

PANDURRA PETROGRAPHIC SAMPLESMoonabie Formation (2)

P.95 and P.96

Corunna Conglomerate (5)

P.90, P.91, P.92, P.93 and P.94

Gawler Range Volcanics (1)

P.24

Roopena Volcanics (1)

P.19 Beda. Bore Chip (A9657)

Pandurra Formation - Basal Section (15)

P.11, P.12, P.18, P.20A, P.20B, P.22, P.25, P.49A, P.50B,
P.86, P.87, P.88B, P.89, EX31A and EX31B

Pandurra Formation - Upper Section (13)

P.21, P.31, P.33, P.43, P.48, P.51B, P.53, P.58, P.67, P.83A,
P.84B, EX29-85584 and EX136-97217

"Elizabeth Creek Conglomerate" (7)

P.2, P.3, P.9, P.27, P.29, P.57A and EX88-86545

Woocalla Dolomite (9)

P.28, P.34, P.35, P.52, EX50 (20 m), EX139-97275,
EX143-97342, EX147-97404 and EX149-97427

"Magazine Hill Clay" (4)

P.6A, P.6B, P.81 and P.85

"Yudnapinna Beds" (14)

P.40, P.41A, P.65, P.65A, P.46, EX11-85235, EX12-85271,
EX36-85713, EX37-85739, EX43-85846, EX44-85893, EX96-86645,
EX142-97325 and EX146 (Cont.).

Whyalla Sandstone (6)

P.51, P.56, P.60, P.66, EX137-97229 (transitional to
"Yudnapinna Beds"?), and EX140-97292

000734

Tregolana Shale (2)

EX93-86587 and EX93-86589

Corraberra Sandstone (2)

P.79 and P.80

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000735

MEMORANDUM TO: Mr. P.J. Curtis

OUR REF: DGT/SS

FROM: D.G. Tonkin

DATE: 26th November, 1974

Typed - 10/12/74.

3 Pandurra Samples for Thin & Polished Section

Three drill cuttings samples as per the attached sheet were sent to I.R. Pontifex & Associates.

The samples are :-

EX 88-86545 : "Elizabeth Creek Conglomerate",
57-60 m. Acid volcanic and acid
"plutonic" pebbles. Carbonate
matrix.

EX 136-97217 : Pandurra Formation quartzite.
84-84.5 m. Green quartzite.

EX 146-Panned Concentrate
: "Yudnapinna Beds"?, 30-46 m.
Carbonate-bearing. Black heavy
mineral.

D.G. TONKIN

000736

Prospect 401

PACMINEX PTY. LIMITED

PETROGRAPHIC AND MINERAGRAPHIC EXAMINATION
OF GULLY AND MG 14 AREAS
DRILL CORE SAMPLES
MOUNT GUNSON

PMR 138/75

SYDNEY

September, 1975

P.J. CURTIS

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II	LOCATION MAP OF GULLY DRILL HOLES
III	LOCATION MAP OF MG 14 AREA DRILL HOLES

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1. INTRODUCTION

Specimens from the Gully and MG 14 areas, drill holes were sent to Head Office for petrographic examination of rock types and mineragraphic study to determine abundance of copper sulphides, grain sizes of the sulphides and complexity of composite sulphides. In particular, the abundance of sphalerite and carrollite and their attachments with copper sulphides were observed. The intergrowths of copper sulphides with syngenetic pyrite were of interest. In both prospecting areas the abundance of clay minerals was of particular interest.

2. SUMMARY

2.1 Gully Area

Classification of Sulphides by Various Textural Lithologies

- a) Fine pyrite framboids are replaced by copper sulphides in variable fine carbonate and/or illite and fine quartz bands.

A 9716, A 9724, A 9727, A 9729, A 9732, A 9733, A 9734 and A 9740.

Sulphides vary to 0.08 mm grain size with median varying between 0.01 mm and 0.05 mm depending on the presence of coarse quartz bands for associated coarser sulphide sizes.

- b) As for (a) but with the additional presence of coarser dense bands of framboidal pyrite being replaced by copper sulphides.

A 9718, A 9731, A 9738, A 9739 and A 9741.

Median size of sulphide grains varies between 0.015 mm in the clayey bands, 0.06 mm in the coarser framboids bands and 0.05 mm in silty quartz bands.

- c) Sulphides fill fractures in dolomite breccia.

A 9719, and A 9720.

Variation of sulphide grain sizes are largely 0.01 mm to 0.05 mm and 0.09 mm to 0.1 mm.

- d) Pandurra Quartzite.

Segregations of sulphides in quartzite interstices.

A 9722.

Size variation of sulphide grains, 0.03 mm to 0.5 mm.

Outline Mineragraphy/PetrographyDrill Hole - Gully 123.15 m (A 9716)

The specimen is mainly fine carbonate with some illite and detrital quartz as bands or erratics. Some quartz is sharply angular. Sulphide present is pyrite and is mainly framboidal. Possible detrital (?) pyrite occurs in detrital quartz bands.

27.30 m (A 9717)

Graded beds of subangular to sharply angular fine quartz possibly derived from an adjoining tillite deposit with very little if any reworking. Illite and kaolinite occur, variably abundant at adjoining horizons.

No banded concentration of sulphides occur but they are present scattered throughout, mainly as framboids and individual tiny spheres of pyrite. Traces of chalcocite are seen replacing pyrite. Occasional subangular coarse pyrite grains present may be detrital but sulphide is largely syngenetic.

27.80 m (A 9718)

No graded bedding is seen, but coarser silty quartz bands contain a high concentration of pyrite framboids (0.015 mm) which show internal replacement by bornite > digenite or chalcocite. The chalcocite where it occurs has extended itself by replacement over several framboids to 0.04 mm to 0.08 mm and is coarse enough to be recoverable but there is not much of it, only occasional bands up to 0.3 mm which are largely unreplaced pyrite.

Pyrite present is syngenetic and the copper sulphides are epigenetic (diagenetic). Other gangue minerals present are illite and kaolinite occurring variably with the quartz in 'bands'.

Drill Hole - Gully 2

20.3 m (A 9719)

Lenses of dark dolomite lie in clear dolomite matrix. Coarse sulphides occupy spaces between dolomite lenses. Bornite with associated digenite and replacing chalcocite occur associated with interstitial clear dolomite, size varying between 0.03 mm and 0.15 mm or as coarse blebs (0.08 mm to 0.5 mm) of the sulphides in coarse quartz-dolomite bands. Very fine sulphides (<0.02 mm) occur in dark dolomite lenses.

Most sulphides are recoverable metallurgically and are epigenetic.

20.5 m (A 9720)

Fragments of dolomite breccia are emplaced in a dolomite cement. Sulphides are variable in size and are scattered or are in lenticular aggregates. Scattered sulphides occur as fine chalcocite, bornite and pyrite, range 0.09 mm to 1.0 mm. Some quite coarse chalcocite grains and chalcocite:bornite composites are present as lenticles in open spaces in dolomite. Sulphides are epigenetic where coarse grained and are mostly copper sulphide. Copper sulphides are recoverable.

23.20 m (A 9721)

The green shale present is a mixture of fine quartz, illite > kaolinite and minor carbonate with a little limonite. The purple shales contain very much more limonite but are otherwise mineralogically similar. Pyrite (to 0.05 mm) occurs rarely in silty bands or in darker shaley areas. Tillitic-looking quartz appears in the shale.

Drill Hole - Gully 4

20.55 m (A 9722)

Red quartzite of Pandurra Formation.

Sulphides have penetrated along interstices of quartz grains as chalcocite or bornite (0.03 mm to 0.5 mm). The coarser sulphides (to overall 70 volume percent) are recoverable.

Drill Hole - Gully 5

21.60 m (A 9723)

The specimen consists of fine bands of dolomite, less associated subangular to rounded quartz and a little illite. The bands contain fine pyrite. In alternating dolomite breccia layers (up to 0.1 mm) are coarse dolomite grains and quartz but much finer dolomite grains occur (down to 0.005 mm) associated with irregular epigenetic pyrite grains containing some greigite (?) centres.

22.60 m (A 9724)

The rock type is similar to that seen at 21.60 m with dolomitic shale and scattered angular coarse quartz and dolomite (probable Collenia) breccia fragments. Sulphides as bornite and chalcocite grains occur scattered along bands and vary, 0.005 mm to 0.05 mm, median 0.01 mm. Most occur through replacement of pyrite framboids although some grains of sulphide appear to be distinctly detrital. Most sulphide would not be easily recoverable by flotation.

Drill Hole - Gully 6

30.65 m to 30.70 m (A 9725)

Sandy, angular and subangular quartz bands and scattered quartz grains, carbonate and clay minerals and limonite. Sulphides have a distinctly detrital appearance and are more concentrated in sandy bands where they are subrounded and appear smaller than associated quartz. The sulphides, (0.005 mm to 0.08 mm, median 0.01 mm) are of chalcopyrite grains being replaced by bornite or chalcopyrite alone, but fine pyrite is also present. The fine pyrite is syngenetic and framboidal. One coarse fracture traversing bedding, contains chalcopyrite, is 0.2 mm wide and only this sulphide would be easily recoverable by flotation after grinding.

Sandy bands show slump and overlying silty bands are microfaulted

31.3 m (A 9726)

Thin layers of silty quartz associated with kaolinite and carbonate alternate with very fine clay mineral/carbonate layers. Fine limonite lenticles occur between the layers. Sulphides present are mainly fine pyrite, and goethite of similar size occurs. The goethite appears to be largely oxidized detrital sulphides.

Drill Hole - Gully 7

28.80 m (A 9727)

A very finely banded shale with bands of angular > rounded quartz (median 0.02 mm) appears as outwash sedimented tillite material; the lamellae grains are parallel to the bedding. Sulphides occur as fine pyrite (median 0.01 mm) with rare replacement by bornite or chalcopyrite.

29.50 m (A 9728)

Quartz pebbles of variable shape and size occur in an abundant clay mineral matrix. Traces only of pyrite occur as a narrow selvage to a quartz pebble or as fine scattered 'fragments'.

Drill Hole - Gully 8

29.50 m (A 9729)

Fine laminated grey shale (quartz, illite and kaolinite) showing crenulations. Quartz grains to 0.05 mm occur scattered throughout or concentrated in silt-sized bands. Sulphides present are mainly pyrite as angular grains or as round framboids. Bornite replaces part of pyrite in some framboids but very little actual copper sulphide occurs.

30.80 m (A 9730)

Quartz porphyrotopes, carbonate grains and rutile:limonite masses occur in a fine groundmass of kaolinite and illite. Goethite is seen as fine, fragmentary and subrounded forms.

Drill Hole - Gully 934.40 m (A 9731)

Coarse, angular (some splintery) and some subrounded quartz grains occur to 0.08 mm in bands associated with occasional large chert and carbonate grains, alternate with illite-rich laminae. All bands are flanked by fine limonite-rich laminae. One coarse (several millimetres wide) pyrite band consists of diagenetically cemented (by secondary pyrite) framboids. The band has a sharp contact with flanking shale minerals. Scattered, large (approx. 0.05 mm) bornite, chalcocite and composites of the minerals occur. Within the quartz bands are subrounded and angular chalcocite (0.05 mm) slightly smaller than accompanying quartz, are of similar shape and are possibly detrital (?) pseudomorphs. In the more shaley bands occur fine pyrite, chalcocite and bornite grains, around 0.006 mm.

The coarse sulphides associated with quartz may be recoverable.

Drill Hole - Gully 1129.65 m (A 9732)

Bands up to 1 mm wide occur of very fine quartz and sulphide (approx. 0.005 mm or less) in a groundmass of illite and kaolinite. Coarser bands of quartz (0.08 mm) contain coarse pyrite grains which are possibly detrital and have suffered some diagenetic growth or replacement of pyrite by copper sulphides. However, copper sulphides present are rarely above the median size of 0.005 mm and so are largely too fine for recovery or too scarce as large grains to be worthwhile recovering.

Drill Hole - Gully 12

25.0 m (A 9733)

Scattered fine quartz occurs in a matrix of clay minerals, scattered pyrite grains (approx. 0.005 mm, with a little to 0.01 mm) and traces of replacive chalcocite.

Angular splintery quartz present suggests derivation from tillite outwash.

Drill Hole - Gully 13

27.5 m to 29.0 m ? (A 9734)

Large quartz as sharply angular to rounded grains (to 0.1 mm) occurs as bands in fine quartz, illite and kaolinite. As is usual the coarser sulphides occur in the silty bands. Sulphides are pyrite > bornite > chalcopyrite and are scattered, with a median size 0.008 mm, some grains up to 0.08 mm. Copper sulphides are scarce and mainly of too fine grain for flotation.

32.8 m to 32.85 m (A 9735)

Sandy bands of coarse and fine subrounded quartz grains merge outwardly into fine quartz and clay mineral. A little coarse carbonate and potash feldspar occur.

Pyrite lenticles contain abundant inclusions of fine angular quartz. Elsewhere, sulphides including some chalcopyrite and bornite with less chalcocite are largely angular grains, median 0.08 mm.

The coarse sulphides here are fairly abundant and would be recoverable by flotation after moderate grinding.

34.5 m (A 9736)

Fine bands of kaolinite > illite and quartz alternate with silty and sandy quartz bands containing interstitial clay minerals and limonite. Scattered, round to angular sulphides (0.015 mm to 0.08 mm) occur in the silty bands. Overall size range of sulphides is 0.015 mm to 0.18 mm. The

sulphides occur as bornite or bornite with pyrite relicts or pyrite alone. Shapes of grains suggest a detrital origin. A little chalcocite also occurs. Copper sulphides are not abundant although they are of suitable grain size for reasonable recovery.

Drill Hole - Gully 14

23.2 m (A 9737)

A 'dyke' of angular to subangular quartz has penetrated a vertical fracture in fine angular quartz, kaolinite and illite beds. Some lateral penetration by 'dyke' material along angular quartz bands occurs.

A large pebble of pyrite (approx. 1 mm) occurs in clayey bands. Other scattered pyrite grains (approx. 0.005 mm) occur but most opaque mineral present is goethite.

24.30 m (A 9738)

Finely laminated shale with a sulphide-rich band of framboidal pyrite (median 0.02 mm) shows some internal bornite > chalcocite replacement. Away from the sulphide band occurs subangular and rounded bornite, chalcocite and less pyrite (median 0.05 mm) which are mainly associated with silty quartz bands. There is some recoverable bornite and chalcocite of suitable grain sizes but the copper sulphides are not abundant here.

24.40 m (A 9739)

Silty quartz bands in shale contain fine parallel laminae of limonite. Scattered pyrite and fewer bornite grains show a maximum size of 0.05 mm and median of 0.008 mm. A lenticle of sulphide which was originally a concentration of syngenetic pyrite framboids shows some replacement by massive bornite and less chalcocite but both sulphides contain abundant pyrite relicts, many from 0.002 mm to 0.005 mm. Elsewhere fine veinlets of bornite (< 0.02 mm) surround pyrite framboids (0.01 mm to 0.05 mm) in aggregates.

Supposing at this level there is much of this lenticular copper sulphide finely intergrown with pyrite it would not be an economic proposition to separate the pyrite.

Drill Hole - Gully 15

21.80 m (A 9740)

Shaley clay and silty clay bands alternate and are better defined and separated by fine limonite-rich parallel laminae. The silty bands are packed with fine angular quartz. Sulphide lenticles as at DH Gully 14, 24.40 m occur, but to a less extent, very fine spheres of pyrite remain in essentially chalcocite pseudomorphs of pyrite framboids. Scattered chalcocite which have pseudomorphed pyrite grains vary to 0.10 mm. Much of the chalcocite is recoverable at this level.

24.50 m (A 9741)

Silty clay and clay bands alternate. Much of the sulphide is around 0.008 mm but varies mainly to 0.05 mm, in places in lenticles to 0.4 mm; and an association with pyrite is seen in many grains. The copper sulphides replacing pyrite are chalcocite > bornite.

A proportion of copper sulphides may be coarse enough and sufficiently free from pyrite to be recoverable here.

Most copper sulphides present are very fine or are intimately associated with pyrite. Milling and flotation would be difficult with fine pyrite and abundant clay associations. Only in limited drill holes where sulphides occur with coarse quartz and carbonate would appreciable sulphides be easily recoverable. The greatest volume of sulphides are commonly in the quartzose silty bands. Some sulphide appears to be detrital although most pyrite is syngenetic and copper sulphides are apparently epigenetic (diagenetic).

Many quartz grains appear fragmented (including much of the finer material) and some tillite horizons of derived outwash material are suspected to be present.

2.2 MG 14 Area

Classification of Sulphides by Various Textural Lithologies

- a) Dolomite Breccia. A 9705, A 9706 and A 9707.
Dispersed sulphide variations, 0.005 mm to 0.27 mm, median sizes vary from 0.01 mm to 0.04 mm. Fracture fill sulphides vary to 0.8 mm.
- b) Clay Mineral and Fine Quartz Bands. A 9715.
Size variation of sulphides 0.003 mm to 0.015 mm, median 0.005 mm.
- c) Clay Mineral, Calcite and Quartz Bands. A 9708 and A 9713.
Sulphides vary to 0.03 mm, mainly median 0.015 mm but some grains vary to 0.08 mm in silty bands.
- d) Sandy Dolarenite. A 9709, A 9710, A 9711 and A 9712.
Sulphide occupies interstices of dolomite and quartz grains. Sulphide grain size variation is mainly 0.02 to 0.36 mm but fewer grains vary from 0.005 to 0.02 mm.

Outline Mineragraphy/Petrography

Drill Hole - MG 62

41.20 m (A 9705)

The rock described as dololutite appears to be Collenia breccia fragments with secondary carbonate cement. Fill and replacement sulphides of bornite and carrollite to 0.80 mm occur. Fine sulphides of bornite and chalcocite are mainly around 0.01 mm. Syngenetic pyrite spheres occur as relicts within the fine copper sulphides. Sphalerite selvages around bornite are seen.

Coarse sulphides may easily be recoverable metallurgically but sphalerite selvages to bornite and association of bornite with abundant carrollite enhance the penalty bearing cobalt and zinc component of final concentrate.

42.0 m (A 9706)

Very similar to 41.20 m in rock type (Collenia dolomite breccia recemented by carbonate) but the larger sulphide grains (to 0.80 mm) are largely of chalcopryrite or chalcopryrite:carrollite composites. Very fine sulphides (median 0.01 mm) are of bornite, chalcopryrite or of composites of these minerals with chalcocite. Pyrite spheres occur in fine sulphides. Sphalerite forms narrow selvages around chalcopryrite. The coarse sulphide would be recoverable but obtaining a pure copper concentrate may be difficult.

Drill Hole - MG 63

39.60 m (A 9714)

Typical basal Whyalla Sandstone as normally occurs stratigraphically immediately above Woocalla Dolomite, with round to subrounded sand size quartz grains; Collenia dolomite fragments and a matrix of fine silt size quartz. Traces of chalcopryrite occur filling a radial fissure in a large quartz grain and a little occurs cementing silt size quartz in the matrix.

39.90 m (A 9715)

A shale rock of fine kaolinite, illite and coarse quartz porphyrotopes. A little carbonate also occurs. Sulphides present are abundant and roughly banded, vary 0.003 mm to 0.015 mm, median 0.005 mm, are of bornite > chalcocite and residual pyrite framboids part replaced by the copper sulphides.

Sulphides are in the main too fine for flotation.

Drill Hole - MG 6433.30 m (A 9707)

An intraformational dolomite breccia. Irregular sized sulphides occur varying between 0.005 mm to 0.27 mm, median 0.035 mm. The sulphides present are bornite and carrollite in the coarser sizes; bornite, carrollite, chalcocite and sphalerite as the finer grains. Many sulphides are fine and complex locked.

No worthwhile copper concentrate could be prepared from this ore, free from cobalt or zinc.

33.35 m (A 9708)

Similar to DH MG 63, 39.90 m rock type (A 9715) above in fine grain size but A 9708 contains abundant fine calcite as well as clay minerals. Scattered fine bornite > chalcocite grains are seen to be replacing framboidal pyrite, median 0.015 mm, maximum 0.03 mm. A little bornite:chalcopyrite and chalcocite occur as free grains. Probably epigenetic replacement by copper sulphides of syngenetic framboidal pyrite has occurred. Rather too fine for efficient flotation.

38.00 m (A 9710)

The disturbance of Collenia dolomite bands was followed by fill between the bands from sand size quartz grains subsequently part-silicified. Sulphides occur patchily as medium to coarse grains (0.09 to 0.3 mm) in the matrix between dolomite fragments. The sulphide present is mainly chalcopyrite and many grains have narrow selvages of sphalerite.

Metallurgical recovery of the chalcopyrite would bring in abundant sphalerite as attachments. A little framboidal pyrite occurs.

Drill Hole - MG 6530.25 m (A 9709)

A sandy dolarenite which consists of oncoliths and ooliths of dolomite and scattered quartz sand grains. Scattered large (0.09 mm to 0.36 mm) complex sulphide composites of bornite and sphalerite and also bornite and carrollite occur. Smaller bornite:chalcopyrite composites also occur which are free from gangue composites and very fine (0.005 mm to 0.02 mm) grains of bornite and chalcocite are seen.

Much of the middle size bornite and chalcocite could possibly be recovered free from carrollite and sphalerite.

30.60 m (A 9711)

Mainly a sandy dolarenite similar to A 9709 above, but slightly finer grained and with quartz free dolomite bands.

Coarse sulphides (including lenticles) occur in coarser dolomite layers, the sulphides varying from 0.02 mm to 0.25 mm are mainly of bornite or bornite complex locked with carrollite or chalcocite. Sphalerite occurs as selvages around bornite grains. This is millable ore, but attachments with sphalerite and carrollite are common.

Very fine interstitial sulphides (0.005 mm to 0.01 mm) occur scattered around as chalcocite, bornite or composites of these minerals. Pyrite framboids also occur. Copper sulphides are too fine for efficient milling and flotation.

36.0 m (A 9712)

Dolomite algal lenticular forms (Collenia ?) and sandy quartz occur with alternating bands of dolomite (dolarenite).

Irregular sulphide (to 0.4 mm) occurs as chalcopyrite with some sphalerite at edges of chalcopyrite crystal masses in interstices of dolarenite bands. Smaller bornite grains occur (approximately 0.05 mm).

In finer interstitial bands, pyrite is replaced by bornite or chalcocite and bornite:chalcopyrite composites occur (mainly 0.03 mm but as fine as 0.005 mm). Very little coarse copper sulphide occurs here.

36.9 m (A 9713)

Silty quartz and illite flakes in a carbonate matrix and fine carbonate with dolomitic Collenia forms occur as alternate lenticles.

Sulphides are largely irregular pyrite framboid groups of spheres showing 'matrix' replacement by bornite or less so by chalcocite. Occasional sub-angular grains of bornite and chalcocite grains (0.03 to 0.08 mm) are seen.

As for the Gully area, pyrite occurs as fine syngenetic framboidal material and is being replaced by bornite or chalcocite in the fine shaley members of the sequence. However, in dolomite breccias and dolarenite rocks the coarser sulphides of chalcopyrite, bornite and occasionally chalcocite are complex locked with carrollite and sphalerite to make reasonable separation of cobalt and zinc from the copper sulphides difficult. Carrollite appears to occur more abundantly with the coarser sulphides than was seen in most Cattle Grid ores.

shale

coarse
seds.

000755

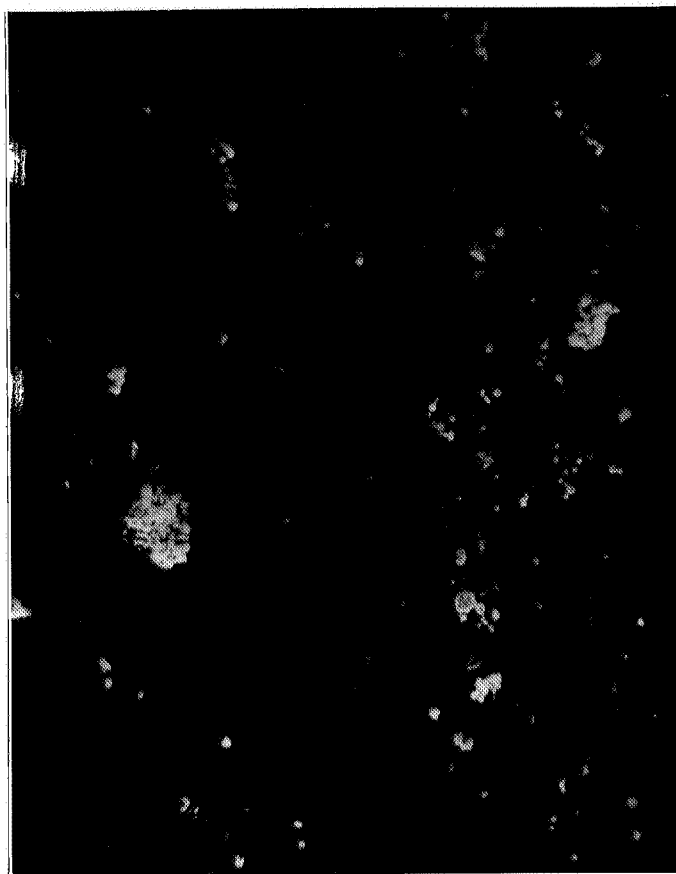


Figure 1 DH Gully 1, 23.15 m (A9716)

Framboidal pyrite grains (raspberry appearance)
in quartz-illite bands. Some framboids have
grey (goethite) centres.

Plain light

Magnification X 116

3. PETROGRAPHIC/MINERAGRAPHIC INVESTIGATION

3.1 Gully Area

Drill Hole - Gully 1

23.15 m (A 9716). Shale:dolomite bands. (Figure 1).

Shale and dolomite bands showing slumping, scour and load casts.

T.S. Shaley and dolomite bands alternate with some gradation in fine bands (to 0.03 mm) and sharp contacts among coarser bands (up to several millimetres wide).

The dolomite bands show relicts of scattered rounded algal forms (approx. 0.02 mm to 0.1 mm). Dolomite grains average around 0.08 mm. Scattered, elongated quartz grains (to 0.08 mm) occur. Dispersed opaque minerals and bands of opaque minerals occur (0.002 mm to 0.03 mm, median 0.007 mm). In the bands they occur as lenticular shapes, and are associated with some detrital quartz of rounded to subangular shapes (median 0.015 mm) and also sparse flakes of detrital illite (approx. 0.02 mm). In the shale bands, opaque minerals occur as single grains (median 0.03 mm) or as lenticles in bands; abundant subangular to sub-rounded quartz grains (approx. 0.02 mm) also occur as well as a little illite and abundant fine carbonate.

Erratics of quartz to 0.1 mm occur as sharply angular or round grains. Similarly, erratics of opaque minerals rather rounded and of around 0.08 mm occur.

P.S. Very finely banded framboids (median 0.007 mm) and single spheres of pyrite (0.002 mm to 0.003 mm) or stringers of pyrite spheres occur along bedding. Coarser

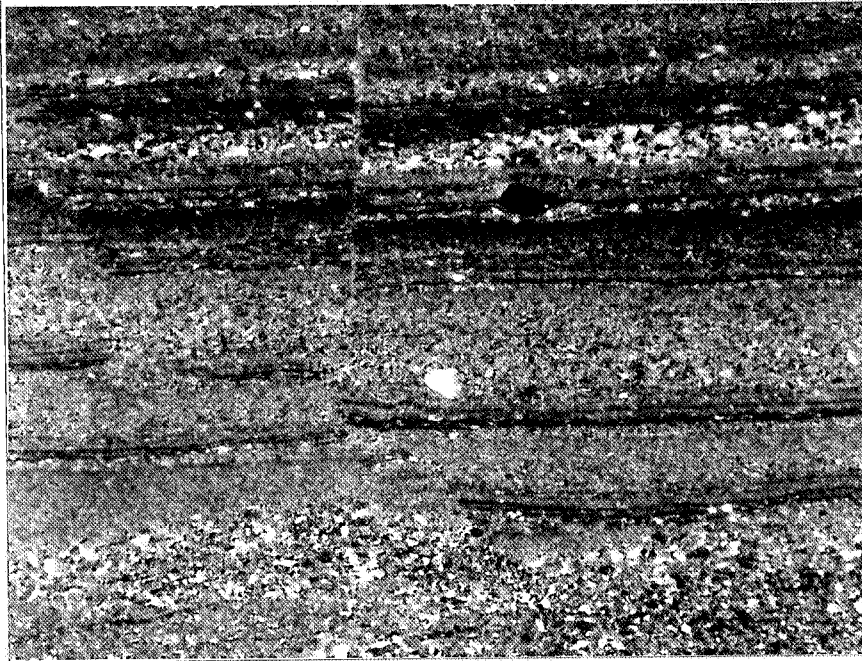


Figure 2 DH Gully 1, 27.30 m (A9717)

Faulted silt and clay bands showing graded bedding. Subrounded detrital grains of mainly quartz are seen.

Crossed nicols

Magnification X 15

pyrite euhedra around 0.01 mm to 0.03 mm are scarce. Some framboids are altering to goethite at centres.

The mass of sulphide (all pyrite) is mainly syngenetic but coarser pyrite grains occur with detrital quartz and may also be detrital.

27.30 m (A 9717). Laminated shale. (Figure 2).

Fine parallel shaley bands occur with silty intercalations of greater thickness and alternating with the dark shale. Some microfaulting of the bands has occurred.

Graded bedding of silty bands of angular to subangular quartz grains (median 0.005 mm top, to median 0.06 mm bottom) is seen. Illite and kaolinite make up the bulk of interstices with scattered opaques, mainly around 0.005 mm.

In the finer silty bands occur fine opaque horizons or laminae (approx. 0.005 mm wide); several occur in parallel formation 0.01 mm apart. Quartz erratics mainly 0.08 mm but up to 0.3 mm occur.

In the coarser silty bands occur scattered opaques (0.005 mm to 0.06 mm) to a density of 10 volume percent.

Much of the coarse silty quartz is very angular and splintery and may be of tillite origin.

Microfaulting is sharp and therefore occurred after consolidation of bands.

P.S. Pyrite framboids (0.005 mm to 0.008 mm) and specks of pyrite spheres (approx. 0.002 mm) occur in all bands but mainly in the quartz bands. Coarse quartz bands show angular grains (approx. 0.035 mm). Bands of the angular detrital quartz and minor goethite (0.05 mm to 0.1 mm) with some breaching of bands occurs.

Traces of chalcocite replacing pyrite are seen in one angular fragment (0.04 mm) and smaller angular pyrite grains also occur rarely or are seen replacing detrital quartz.

Syngenetic pyrite is seen in all bands but much of it occurs in quartz rich bands. Possibly a little detrital pyrite also occurs associated with quartz. No evidence of framboidal pyrite association is seen in the detrital grains. Many framboids are angular like the quartz grains but are appreciably smaller (to half size) compared with associated angular quartz grains.

27.80 m (A 9718). Laminated Shale.

Finely laminated and fissile. Microfaulting of selective bands occurred after consolidation resulting in unfaulted bands resting on faulted layers.

T.S. Graded bedding in bands is not noticable here as in the section from 27.30 m, but coarser silty bands have a high concentration of opaques (10 to 15 percent) at the centre of one fine silt band (1.7 mm wide). The opaques vary, 0.005 mm to 0.03 mm. Subangular quartz (approx. 0.01 mm) occurs associated with opaques and about half of non-opaque mineral is mainly kaolinite > illite. The fine repeated (approx. 0.005 mm wide) banding of limonite/bituminous matter occur as parallel laminae and are associated with quartz (approx. 0.01 mm), illite, kaolinite; but also erratics of larger rounded quartz grains (approx. 0.05 mm) occur.

The quartz in all bands is variable from very angular to round grains. Some of the sharply angular grains may be outwash (?) from a neighbouring tillite formation.

The regular parallel banding indicates quiet sedimentation in a lagoonal environment. The larger erratics indicate that the shoreline was not far away. The erratics have made no impression on the bedding thus confirming quiet deposition. Faulting occurred after consolidation of individual bands which indicate drying out periods of lagoon (?).

P.S. Similar to 27.30 m (A 9717) but bands of framboidal pyrite associated with quartz grains are rounded; framboids are of median size, 0.015 mm. Much of the interiors of framboid particles are replaced by bornite > digenite. Where it occurs chalcocite has replaced much of the pyrite in the framboids and has extended itself over two to three adjoining framboids (to grain sizes 0.04 mm to 0.08 mm) but these enlarged copper sulphide masses are less common. Where bornite or bornite and digenite replace pyrite in framboid spheres, normally 50 to 70 percent of pyrite grains remain as radiate pyrite slivers.

Coarse chalcocite (the result of replacement of pyrite framboids) is probably recoverable.

Drill Hole - Gully 2

20.3 m (A 9719). Dolomitic black shale.

Lenses of dark dolomite occur in clear matrix. Sulphides occupy spaces between dolomite lenses and appear in hand specimen to be epigenetic.

T.S. Darker dolomite breccia fragments contain some fine opaque limonite and average 0.6 mm, occur in a coarser clear irregular dolomite cement. Some lenticles of bedded dolomite (to several millimetres in length) occur brecciated. Sulphides occur as irregular veins and masses (0.1 mm to 1.0 mm) either replacing dolomite lenticles or the clear dolomite matrix cement.

P.S. Bornite with digenite are associated with replacing chalcocite and occur in fractures (0.03 mm to 0.15 mm) associated with scattered large pyrite (approx. 0.06 mm) dolomite and quartz grains. Also coarse blebs (0.08 mm to 0.5 mm) of chalcocite with bornite occur in the interstices of narrow coarse quartz: dolomite bands. Very fine (< 0.02 mm) chalcocite and bornite grains occur scattered through the fine shaley areas.

The coarser sulphides equal >90 percent of sulphides present but are scattered unevenly through the rock matrix. These coarser sulphides are probably recoverable.

20.5 m (A 9720). Dolomite/shale bands with bornite.

Thin carbonate bands are seen.

T.S. Fragments of dolomite (dolomite breccia) occur in a dolomite cement. Sulphides up to 0.5 mm have marginally intergrown with quartz (approx. 0.06 mm) in mosaics and are associated with recrystallized rhombs of dolomite. The sulphides have recrystallized in open vugs lined by dolomite and quartz to form cement of rock fragments. Some part replacement of dolomite fragments by sulphides occurs.

P.S. Scattered fine chalcocite, bornite grains and pyrite framboids (0.01 mm to 0.05 mm) occur but larger lenticles of bornite:chalcocite intergrowths (0.09 mm to 1.0 mm) are seen which are probably replacements of bornite (associated with some digenite) by chalcocite. Occasionally these coarser sulphides nucleate around large quartz grains and mosaics. The T.S. evidence indicates some replacement of dolomite and quartz by epigenetic (diagenetic) sulphides.

The coarser lenticles of sulphides are recoverable and these constitute about 70 percent of sulphides

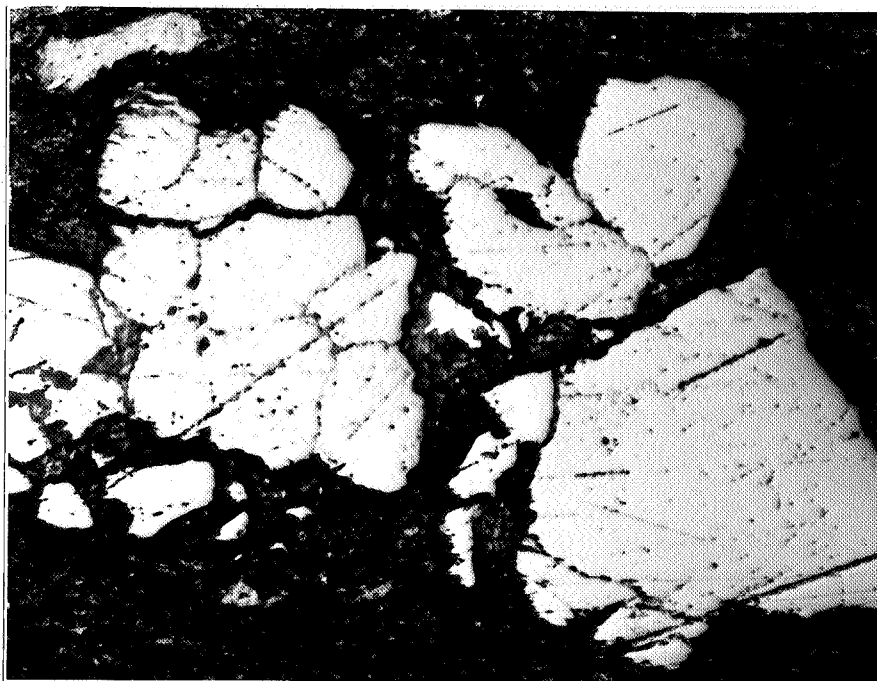


Figure 3 DH Gully 2, 23.20 m (A9721)

Coarse goethite (off-white) with a small chalcocite grain (pure white) at centre of photograph. Surrounded by mainly fine illite and quartz in bands.

Plain light

Magnification x 80

present. The finer disseminated sulphides are too fine for an acceptable recovery ratio.

23.20 m (A 9721). Brown/green shale. (Figure 3).

Sharply differentiated green and purple shale. Both types contain silty laminae.

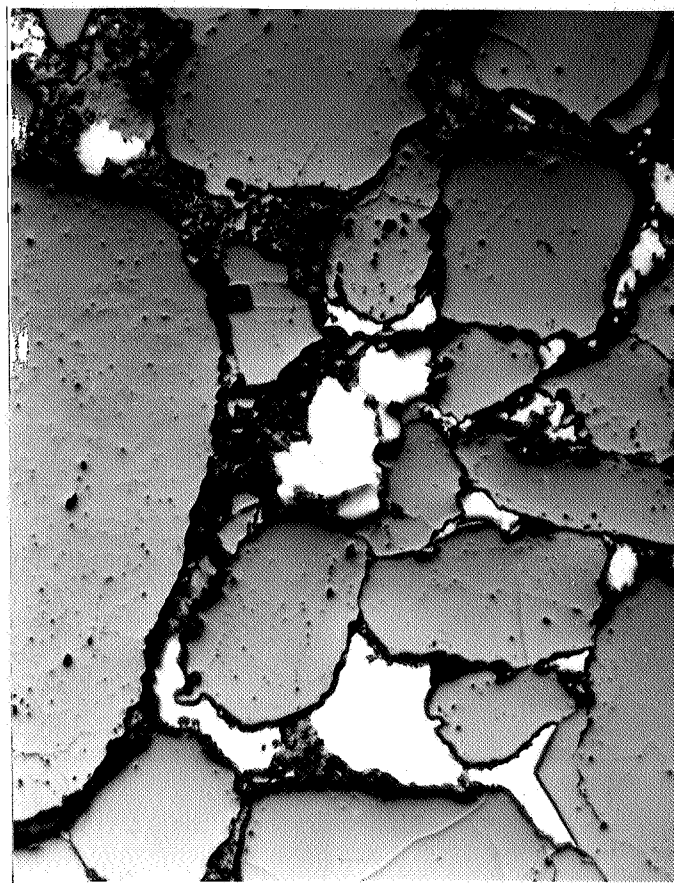
T.S. Shows some gradation between the green and purple fine shale. Bands and lenticles up to 1 mm wide of subangular to rounded quartz (0.05 mm to 1 mm) of very variable sizes occur within the bands with an interstices of the fine shaley minerals as occur in the bedded shale. Some bedding of fine silty quartz (down to 0.005 mm) occurs in both shale types.

In detail the green shale is seen to contain angular to subangular quartz (median 0.01 mm) > illite > kaolinite, scattered carbonate grains (approx. 0.01 mm) and appreciable opaques as fine laminae and subangular grains (approx. 0.04 mm). Sharply angular quartz erratics (to 0.2 mm) occur in bands (occasional silty bands).

The purple shale contains abundant round to angular quartz, often as sharp sliver like grains up to 0.2 mm long; abundant limonitic opaques of similar sizes and shapes; and illite > kaolinite. Coarser detrital quartz bands are largely cemented by coarse opaque minerals.

Abundant sulphide appears to be syngenetic with the shale yet much of it appears to be detrital. (See Binda - Reference 1). The grains are rounded or jagged of similar shape but smaller size to quartz grains with which they are bedded.

P.S. Pyrite occurs to 0.05 mm in coarse quartz bands (0.5 mm to 3.0 mm wide). Traces of very fine pyrite occurs in the darker shaley areas.



0.2 mm

Figure 4 DH Gully 4, 20.55 m (A9722)

Main Open Cut area, Pandurra Formation.

Chalcocite (pure white) occurs with clay mineral (dark) in the interstices of quartz grains (pale grey).

Plain light

Magnification X 83

Drill Hole - Gully 4

20.55 m (A 9722). M.O.C. area, P.Q. (Figure 4).

Large (several millimetres in size) red quartzite fragments occur in pink quartzite matrix. Sulphides occur sparsely in red quartzite fragments replacing matrix between quartz grains but are common in the interstices and filling vugs of pink quartzite, particularly at the contact between red and pink quartzite.

T.S. Red quartzite shows mainly rounded quartz grains (0.7 mm to 1.5 mm) with authigenic quartz overgrowths so that subsequent epigenetic sulphides have not penetrated except along minimal channelways presently occupied by clay.

Pink quartzite is loosely compacted with smaller quartz grains (approx. 0.15 mm) and large quartz erratics (approx. 0.7 mm). More clay matrix and greater penetration of sulphides (0.03 mm by 0.15 mm wide) has taken place, as veins along the channelways. Very little replacement of quartz by sulphides is seen.

P.S. Chalcocite or chalcocite replacing bornite (0.03 mm to 0.5 mm) occurs filling the interstices between large quartz grains. The sulphide occurs to 5 volume percent and is largely recoverable. However, a little of the sulphide is finely interstitial (approx. 0.03 mm wide) and accounts for about 30 percent of sulphides present.

Drill Hole - Gully 5

21.60 m (A 9723). Dolomite/shale bands.

Fine, black shale bands alternate with fine dolomite breccia bands.



0.1 mm

Figure 5 DH Gully 5, 22.60 m (A9724)

Framboidal pyrite grains (raspberry forms) are being replaced at centres by bornite (pink). Bornite grains occur with few remaining pyrite spheres. A pyrite, chalcocite (blue) lenticle is seen, (to the right and above centre of photograph).

Plain light

Magnification X 240

T.S. Very fine bands of dolomite > subangular to rounded quartz (median 0.014 mm) > illite (0.02 mm x 0.002 mm) occur. Fine opaques (syngenetic or detrital), < 0.02 mm are disseminated in this layer to 5 volume percent.

In some coarser dolomite breccia layers, carbonate up to 0.1 mm but commonly much finer is the only non-sulphide mineral present. Elsewhere coarser quartz particles (to 0.3 mm) subangular to rounded occur in the dolomite breccia layers. Sulphides present here are fairly coarse to fine (0.1 mm down to 0.005 mm).

P.S. Pyrite occurs as irregular masses (< 0.02 mm to 0.8 mm) distributed along bands with or without coarse irregular quartz. Most pyrite is in the dolomite/quartz bands. The pyrite masses enclose some fine dolomite grains and they appear to have grown through diagenetic addition to original small syngenetic pyrite grains. Some very small (approx. 0.02 mm) equally irregular pyrite grains occur more sparsely in the shaley bands.

22.60 m (A 9724). Dolomite shale bands. (Figure 5).

Fine dark bands alternate with fine intraformational dolomite breccia bands.

T.S. As at 21.60 m (A 9723), fine shaley carbonate of dolomite > angular to rounded quartz (< 0.005 mm to 0.03 mm) > fine illite. Quartz rich bands in this layer appear to contain most of the sulphide (range 0.005 mm to 0.05 mm).

In the dolomite layers of fine intraformational breccia, quartz erratics occur to 0.07 mm, sulphides are scattered and vary from 0.005 mm to 0.06 mm. Most abundant sulphides range is between 0.005 mm to 0.01 mm.



Figure 6 DH Gully 6, 30.65 m to 30.70 m (A 9725).

Sandy bands of quartz grains (white) mixed with fine carbonate and clay minerals show recumbent slump folds. Underlying finer silty carbonaceous bands of fine quartz and kaolinite show microfaulting.

The lower faulted silty bands are the top of a preceding sandy-clay to silty-clay sequence and a sharp contact between the silty layers and the overlying (disturbed sandy) bottom of the next sequence is observed.

Plain light

Magnification x 15

There is a sharp contact between the dolomite 'breccia' and shaley bands.

P.S. Bornite and chalcocite occur as somewhat angular to rounded grains in the angular quartz bands. The sulphide grains range from 0.005 mm to 0.05 mm, median 0.01 mm. Some sulphide grains show diagenetic addition of chalcocite, others are subrounded to subangular and are of similar or slightly smaller size than the quartz grains with which they are associated. One other bornite grain shows a quartz inclusion embedded at the periphery which does not alter the subrounded outline of the grain. Yet other bornite grains are speckled with fine relict pyrite spheres and are replacement type composites. Bornite replaces the interior areas of some pyrite framboids.

Probably much of the bornite and chalcocite are the result of replacement of pyrite. Yet other grains have a distinctly detrital appearance.

Very little of this sulphide is recoverable from a metallurgical point of view, it is mainly too fine.

Drill Hole - Gully 6

30.65 m to 30.70 m (A 9725). Laminated shale and arenaceous dolomite bands. (Figure 6).

Sandy dolomite and dolomitic shale bands.

T.S. In slumped sandy bands, angular to subangular quartz (0.005 mm to 0.15 mm, median 0.045 mm) occurs to 20 volume percent, but is near 80 percent in packed laminae of quartz grains (approx. 0.5 mm x 0.1 mm); carbonate > kaolinite > illite also occur. Fine laminae of limonite occur which enhance the bedding planes. Sulphides occur to 5 to 7 percent (0.005 mm to 0.08 mm) and are apparently mainly detrital. A little

greenish clay mineral also occurs. Part of sulphide (0.2 mm wide) occurs as a 'vein' flanked by zones of diagenetic carbonate.

The sandy bands grade into finer bands of silty quartz (0.02 mm) as subangular quartz in a groundmass of kaolinite. Some scattered carbonate granules (approx. 0.025 mm) and illite also occurs. Faulting of some silty bands is seen. Syngenetic and detrital (?) sulphide grains (0.01 to 0.025 mm) occur to 2 to 3 volume percent.

P.S. Complex sulphide composites occur with sizes of grains varying from 0.005 mm to 0.08 mm, median 0.01 mm. Much of the sulphide is of chalcopyrite which is seen to be replaced by bornite or of chalcopyrite with a nucleus of bornite. Less chalcocite occurs replacing bornite and some of the finer values are of syngenetic pyrite. Some of the chalcopyrite and bornite grains are sharply angular, others are angular to subrounded which suggests detrital origin for some of the sulphide grains.

Chalcopyrite occurring in a vertical fracture (several millimetres in length and 0.2 mm wide) is encapsulated by a zone of pure carbonate.

Only the sparse chalcopyrite noted in vertical fractures is recoverable. Fine grain size would prevent much of the remainder from responding to milling and flotation.

31.3 m (A 9726). Laminated Shale.

Finely laminated shale, contains scattered silt-size quartz particles.

T.S. Silty quartz (0.005 mm to 0.08 mm) of subangular grains in layers with kaolinite and carbonate. The layers are roughly 0.3 mm wide, alternate with clayey layers of mainly kaolinite > carbonate (approx. 0.005 mm)

and scattered angular to subangular quartz grains (to 0.06 mm). Limonite lenticles (approx. 0.02 mm wide) occur between the layers. Opaques are mainly very fine (0.005 mm to 0.01 mm) and are scattered through all layers, to 2 volume percent.

P.S. Massive pyrite occurs in a lenticle (0.4 mm x 3 mm) with quartz. Elsewhere fine pyrite grains (approx. 0.005 mm) and goethite of similar size occur scattered throughout. Some goethite (pseudomorphs of pyrite ?) is subrounded and appears to be detrital.

Drill Hole - Gully 7

28.80 m (A 9727). Laminated shale.

Very fine banded shale.

T.S. Angular > rounded quartz (0.005 mm to 0.15 mm, median 0.02 mm) varies in abundance from 5 to 20 percent in bands where quartz which occurs as mainly lamellar fragments are parallel to bedding. These are possible outwash grains from till (?).

The sulphides (0.005 mm to 0.06 mm, median 0.01 mm) present are more common (to 5 to 7 percent) in the more quartz-rich bands. The grains are subangular to rounded and are possibly detrital, or maybe diagenetically modified syngenetic. The remaining minerals present are illite flakes > kaolinite.

Fine limonite laminae are close together (0.03 mm to 0.01 mm apart) in the more shaley bands. These laminae are further spaced apart in the silty quartz bands.

P.S. Moderate fine pyrite is seen including many angular grains which occur at around 0.01 mm. Rare bornite: chalcopyrite composites (0.005 mm to 0.01 mm) show

replacement of bornite by chalcopyrite.

29.50 m (A 9728). Green clay and quartz pebbles.

Scattered, sand size quartz clasts occur in a groundmass of greenish clay.

T.S. Rounded to subangular quartz rock fragments and mineral grains (0.01 mm to 2 mm) occur as irregular sizes and shapes. Some quartz rock fragments are silty equigranular quartzite but most are of irregular quartz mosaics. The groundmass is of mixed fine kaolinite and less illite. Some reddish brown iron oxide cherty clasts occur and opaque minerals as scattered grains which are mainly, <0.01 mm but to 0.05 mm and larger occur from 2 to 3 percent.

P.S. Traces only of pyrite (0.005 mm to 0.010 mm) are seen as a selvage to a large quartz grain and as scattered fragmentary grains (0.18 mm x 0.04 mm down to 0.02 mm).

Drill Hole - Gully 8

29.50 m (A 9729). Laminated shale.

Fine laminated grey shale. Crenulation of shaley bands is seen.

T.S. Quartz (up to 0.05 mm) occurs scattered through all shaley bands but is more abundant in silt-sized bands. Microfaulting of shaley bands occurs.

Much of the quartz present is sharply angular, range 0.01 mm to 0.05 mm with most grains being around 0.05 mm. However, some grains are quite rounded. All bands contain quartz, illite and kaolinite but in variable amounts and with transitional variations between the bands. Fine limonite laminae emphasise

the bedding. Sulphides look to be detrital (or syngenetic) and vary <0.005 mm to 0.05 mm but largely are around 0.01 mm occurring to 5 percent of the rock.

Sulphides are fairly abundant in the silty bands (detrital ?) but also occur as laminae in, or are associated with limonitic laminae, (syngenetic ?).

P.S. Much fine pyrite (0.002 mm to 0.07 mm, median 0.005 mm) occurs, some of it present as quite angular grains whereas other grains are rounded framboids or filled-in framboids (with secondary pyrite). Bornite is replacing some of the pyrite grains in places. Much of the pyrite is sharply angular.

Negligible copper sulphide occurs here.

30.80 m (A 9730). Purple silt with pebbles.

Quartz prophyrotopes occur in a fine grey clayey matrix. Abundant specks of 'sulphide' are seen.

T.S. The quartz grains are rock fragments in the larger sizes and mineral grains in the finer sizes (range 0.005 mm to 3 mm with many grains around 0.5 mm). The grains are irregularly distributed and are mainly subangular to rounded. The rock fragments are mainly inequigranular and include occasional quartz gneiss grains, some cherty grains and rutile/limonite masses, (approx. 0.06 mm). Scattered carbonate grains (0.01 mm to 0.02 mm) also occur. Fine opaque minerals (<0.005 mm to 0.02 mm) are densely distributed in the matrix, are fragmentary in appearance and are largely limonitised. The groundmass mineral is largely kaolinite > illite. Some detrital muscovite flakes (approx. 0.1 mm) also occur.

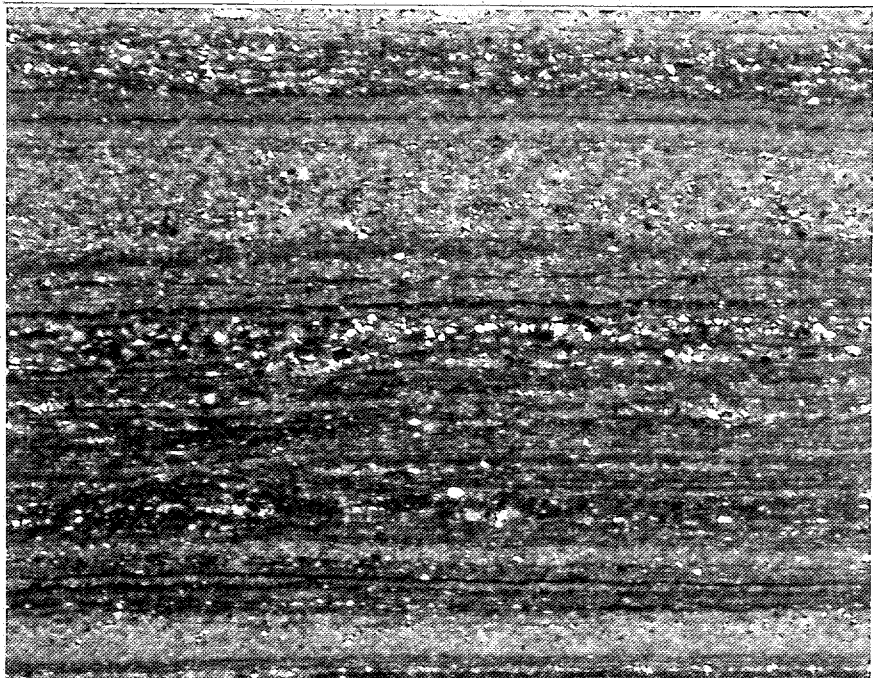


Figure 7 DH Gully 9, 34.40 m (A9731)

Repetitive layers of quartz (white) and sulphides (black) in illite (brown). Limonite bands (dark brown) follow bedding.

Plain light Magnification X 15

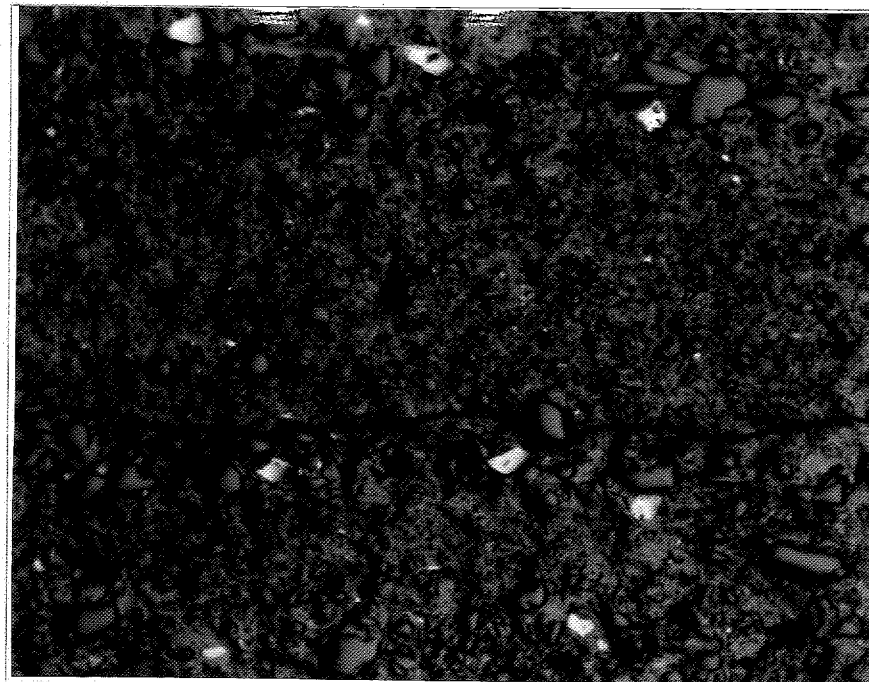


Figure 8 DH Gully 9, 34.40 m (A9731)

Subangular and subrounded bornite grains (pink) and chalcocite (white) are associated with similar shaped but larger detrital quartz grains (brown, dark borders).

Plain light Magnification X 80

P.S. Only goethite is seen (< 0.005 mm to 0.016 mm) as fragmentary and subrounded forms in the matrix.

Drill Hole - Gully 9

34.40 m (A 9731). Laminated shale. (Figures 7 and 8).

Clay and silty quartz bands inter laminate. A coarse (several millimetres wide) band occurs at the base of a siltstone lamina.

T.S. Laminae are made up of limonite lenticles up to 0.005 mm wide which flank illite bands up to 0.05 mm wide. These are interposed with laminae of mainly rounded and subangular quartz up to 0.08 mm but also much smaller. Some sharply angular quartz grains also occur of all sizes, median grain size is around 0.02 mm. Some cherty and carbonate grains occur with quartz but are rare.

Sulphides vary to maximum of 0.08 mm with a median size of 0.005 mm and varying in shape from angular to rounded as does the quartz. In the quartz laminae size and shapes are similar to quartz and similarly for the illitic bands where sulphides are mainly around the median grain size. Many of the sulphides are distinctly detrital in appearance and there are about 3 volume percent present.

P.S. One coarse band of sulphide, mainly pyrite (> 3 mm wide) is differentiated into a solid pyrite band containing occasional inclusions of chalcocite and bornite and some sharply angular quartz of similar size (approx. 0.05 mm) and shape; a wider zone of part filled-in pyrite framboids also containing much detrital quartz and scattered bornite and chalcocite grains (approx. 0.05 mm); another narrow band of dense pyrite is seen with a narrow selvage of sphalerite (?).

Within quartz bands (outside the pyrite band) occur subrounded and angular chalcocite grains (median 0.05 mm) of similar or slightly smaller size to accompanying quartz grains and are of similar shape. Possibly these are pseudomorphs of detrital origin. In the more shaley bands, pyrite, chalcocite and bornite grains are around 0.006 mm.

Sulphides within the quartz bands are recoverable.

Drill Hole - Gully 11

29.65 m (A 9732). Laminated shale.

Laminated shale as for A 9731 but coarser silty, quartz bands.

T.S. Much of the section shows bands up to 1 mm of very fine quartz and sulphide (< 0.005 mm or around 0.005 mm) in a groundmass of illite and kaolinite. The coarse (to 0.06 mm) sulphides in coarse quartz (to approx. 0.08 mm) appear to be diagenetic in their irregular outlines of grains although some large sulphide grains (to 0.05 mm) in the fine bands appear to be detrital and like some quartz grains have disturbed the shale-limonite laminae where they fill in. Diagenetic sulphides cut-across laminae of shale.

There has been local mobilization of sulphides which in the case of coarser grains was possibly largely detrital.

Sharply angular quartz grains occur in some silty bands and their long axes are parallel to the banding.

P.S. Scattered sulphides occur, fairly sparse, and are of pyrite, chalcocite or bornite replacing pyrite. The grains are normally quite angular and vary from

0.006 mm to 0.026 mm, median 0.005 mm. Pyrite varies to 0.06 mm.

Sulphides are mainly too fine for recovery.

Drill Hole - Gully 12

25.0 m (A 9733). Laminated shale.

No chip specimen remains after section preparation.

T.S. The rock consists mainly of fine illite and kaolinite, with scattered fine quartz (10 to 15 volume percent) and sulphides which concentrate into bands (to 20 percent). The quartz is commonly sharply angular and varies from very fine, approx. 0.005 mm to 0.12 mm, and is apparently detrital. The sulphides are also very fine, much of them being in sizes up to 0.005 mm, although overall possibly 10 percent of sulphide present is 0.01 mm. It has a distinctly powdery appearance as fine particles or aggregates of particles and some goethite occurs.

Possible tillite outwash ?

P.S. Sulphides present (approx. 0.005 mm) are pyrite with traces of replacive chalcocite. Appreciable sulphide present is quite angular, but some is lenticular or rounded framboid forms.

Drill Hole - Gully 13

27.5 m to 29.0 ? m (A 9734). Laminated shale.

Pale shale with some fine dark laminae and silty bands.

T.S. Quartz varies in size to 0.1 mm and in shape from sharply angular to rounded. Several large rounded grains are part coated with illite. Much of the quartz is as fine as the illite and kaolinite present.



Figure 9 DH Gully 13, 32.8 to 32.85 m (A9735)

Irregular quartz (white and grey-blue); some associated potash feldspar (twinned - centre of photograph), in carbonate matrix (speckled, pink, buff, etc.). Sulphides present (black) are of similar size to smaller quartz grains.

Crossed nicols

Magnification X 80

Some sulphide (in fine quartzose areas) in zone thicknesses shows long lenticular forms (approx. 0.4 mm x 0.05 mm) and is syngenetic modified by diagenesis. However, much of sulphide (approx. 0.03 mm) present has a distinctly subangular to rounded detrital appearance and is commonly of similar size and shape to associated detrital quartz.

Current scour of beds is common throughout at top-most contact of each band.

P.S. Scattered pyrite occurs; sharply angular and rounded grains are seen. Median grain size is 0.008 mm, size range is (<0.001 mm to 0.08 mm). Scattered bornite grains (0.008 mm to 0.04 mm) with chalcopyrite relicts occur. These are seen to have replaced quartz in places and are irregular in shape, others are almost perfect oval shape.

Some detrital sulphides are present. Too little copper sulphide is present for satisfactory beneficiation.

32.8 m to 32.85 m (A 9735). Laminated shale with sand lenses. (Figure 9).

Sandy lenses occur in shale. Siltstone bands are seen. Diagenetic pyrite nodule, has been transported since a depression has formed in underlying shale band.

T.S. Sandy bands are 3 to 4 mm thick of rough coarse and fine subrounded quartz - two populations occur, median 0.3 mm and median 0.05 mm. Minimal carbonate occurs between the smaller grains and some microcline occurs. Lenses of kaolinite > illite occur within this band.

The sandy band merges upwards into a silty/illite/kaolinite band where quartz has a median size of 0.015 mm, but with subrounded detrital quartz erratics up to 0.1 mm.

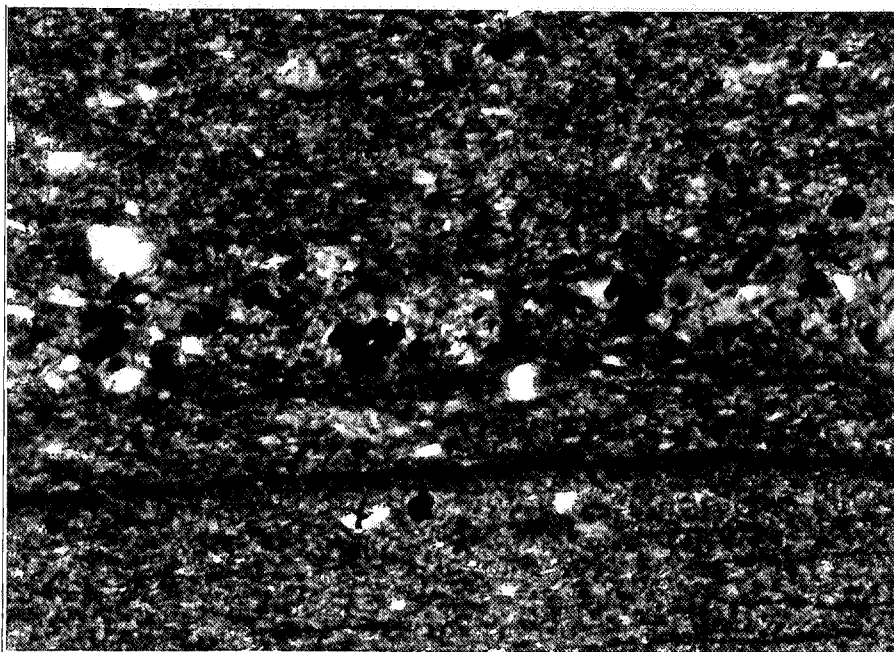


Figure 10 DH Gully 13, 34.5 m (A9736)

Relatively coarse rounded to angular quartz (whites and greys) in bands with smaller rounded and concave bottomed sulphide grains (black). Matrix is fine quartz and clay minerals (yellow-brown).

Crossed nicols

Magnification X 79

Silty bands of mainly subrounded quartz (approx. 0.05 mm and 0.08 mm) occur in illite > kaolinite bands. Some carbonate occurs as isolated grains and fill between quartz and a little potash feldspar.

Sulphides are most common in sandy and silty bands where the grain sizes average 0.05 mm or 0.02 mm respectively (somewhat smaller than associated quartz) and are subrounded or diagenetic, irregular or laminar; some vertical fractures in rock are filled with sulphides. In the shaley bands sulphides occur as detrital erratics, diagenetic irregular grains. Some very fine and dusty (< 0.001 mm) opaque mineral is seen. Active currents were seasonal (?) and scour tops of shaley bands, silt and sand occupy scour hollows. Outwash coarser rock material occurs with seasonal incursion. With bands of silt, the appearance is rather like typical Yudnapinna Beds.

P.S. Several large pyrite lenticles (up to 2.5 mm x 0.5 mm) contain abundant fine angular quartz. Elsewhere sulphides are largely angular. Some chalcopyrite and bornite are seen to be replacing detrital quartz grains, others show fine inclusions of quartz relicts. The sulphides present are bornite > chalcopyrite > chalcocite, some pyrite, and some goethite occurs. The size range is 0.003 mm to 0.10 mm, median 0.08 mm. The sulphides are scattered among larger and similar sized and shaped quartz grains.

The sulphides seen here may respond well to mineral processing by flotation.

34.5 m (A 9736). Laminated shale/silt. (Figure 10).

Interlaminated shale and silt.

T.S. Coarse, rounded > angular quartz and some potash feldspar and chert grains, median 0.08 mm occur fairly equigranular as bands (0.2 mm to 0.8 mm wide) which



Figure 11 DH Gully 14, 23.20 m (A9737)

Detrital quartz (white) in dyke breaks through illite/kaolinite/fine quartz of shale beds (fine speckled brown and white). Pyrite is present (black) and appears to have been displaced upwards (?).

Plain light

Magnification X 15

grade into shale (kaolinite, illite and fine quartz) and somewhat thicker bands containing quartz erratics of sand size as above. Fine limonite lamellae occur in shaley bands, and include some round limonite (after pyrite ?) grains (approx. 0.005 mm).

Scattered round to angular sulphides (0.015 mm to 0.08 mm) occur mainly in the sandy bands and appear to be detrital. Smaller grains (0.015 mm) occur mainly as erratics in the more shaley bands although some larger (approx. 0.08 mm) sulphide grains also occur in the shaley material.

Much of the sulphide here appears to be detrital.

P.S. Similar to preceding A 9735 specimen. Coarse quartz bands are associated with mainly bornite grains, with bornite with pyrite relicts or pyrite alone. The shapes of sulphides, particularly of the pyrite grains suggest a detrital origin. They are mainly somewhat smaller than the accompanying quartz grains in the bands although there are exceptions and some quite large bornite grains occur. A little chalcocite occurs. Range of sulphide grains is 0.02 mm to 0.18 mm, median size 0.07 mm.

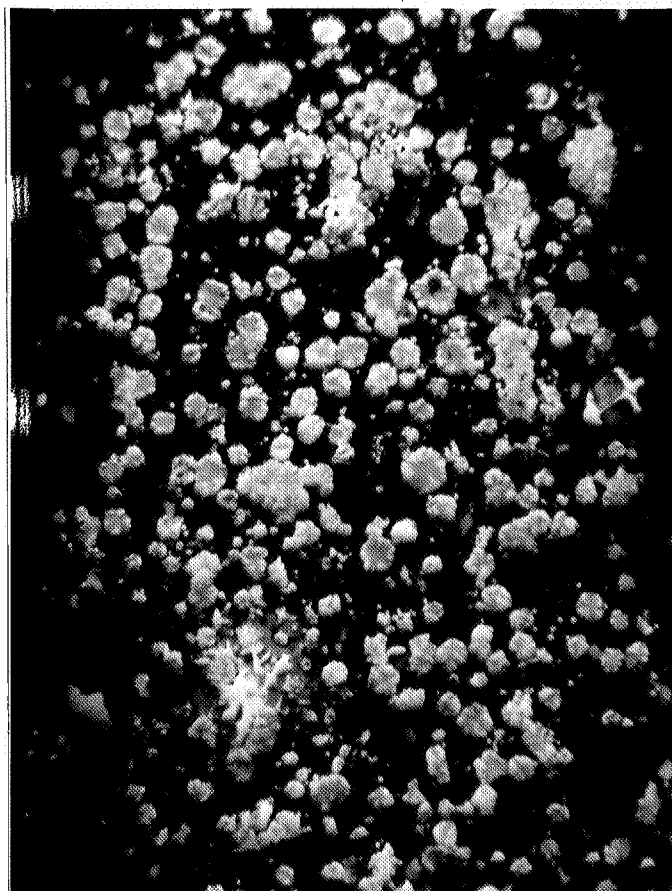
Most of the sulphide is recoverable by flotation but is not abundant.

Drill Hole - Gully 14

23.20 m (A 9737). Laminated shale with mud dyke. (Figure 11).

Siltstone and shale fragments filled a fracture in a moist shale horizon.

The dyke is mainly of angular quartz plus a little microcline of similar size (0.05 mm to 0.8 mm) and shaley



0.2 mm

Figure 12 DH Gully 14, 24.30 m (A9738)

A band of framboidal pyrite grains (yellow) shows replacement by bornite (pink) at centres of grains. Replacement by chalcocite (blue) also occurs. Relict pyrite in grains shows peculiar radial forms.

Magnification X 150

lenticles. Intermixing of sandy material and neighbouring wet shale has probably occurred in the fracture through gentle compaction from above (?). Diagenetic sulphide has filled fractures within the dyke. A little detrital sulphide occurs with the quartz grains of dyke.

Shale beds contain narrow quartz silty bands (0.1 mm to 0.2 mm wide) but these are scarce. Except for local squeezing in of quartz mainly along horizontal bands, some breaching of shale bands occurs. The shaley material is however, largely undisturbed by the intrusion of the dyke. This suggests that a fracture was already present sufficient to accommodate overlying sandy material. Abundant very fine opaque minerals occur in the fine shaley layers. The opaques average about 0.007 mm in size and are angular, of similar size to the very fine angular quartz, illite flakes and kaolinite, which are also present.

Quartz in bands has a very angular aspect.

P.S. One large (1 mm x 0.5 mm) pebble of pyrite occurs. The pebble contains an upper periphery of small quartz grains. A zone of clay minerals occurs between the bottom side and neighbouring quartz silt. The pyrite is a mass of framboids with some interstitial greigite (?).

Elsewhere, occasional pyrite grains (approx. 0.005 mm to 0.015 mm) are seen but most opaque mineral present is fine to large grains of goethite or secondary hematite. Pyrite present is mainly in the 'dyke'.

24.30 m (A 9738). Laminated Shale. (Figure 12).

Finely laminated shale with a sulphide band (approx. 0.7 mm wide).

T.S. The sulphide band consists of aggregates of subangular to subrounded grains of similar size (approx. 0.06 mm)

intermixed with minimal to appreciable clay mineral and few detrital quartz grains. No sharp contact is seen but the band is gradational with overlying shale which contains round to angular detrital quartz mainly around 0.006 mm, smaller fine illite and kaolinite grains. Also narrow dusty brown limonitic bands occur. Erratics of rounded quartz and sulphide (to 0.08 mm) occur in the shaley bands.

P.S. One band parallel to the banding is of framboidal pyrite grains (0.005 mm to 0.05 mm, median 0.02 mm) showing internal replacement by bornite > chalcocite. Curious radial pyrite growths are left in spheres in the pyrite framboids by replacing bornite or chalcocite. Outside the band, subangular and rounded bornite, chalcocite and less pyrite (median 0.05 mm, maximum 0.1 mm) occurs through the groundmass but much of it is confined to narrow quartz bands.

24.40 m (A 9739). Laminated shale.

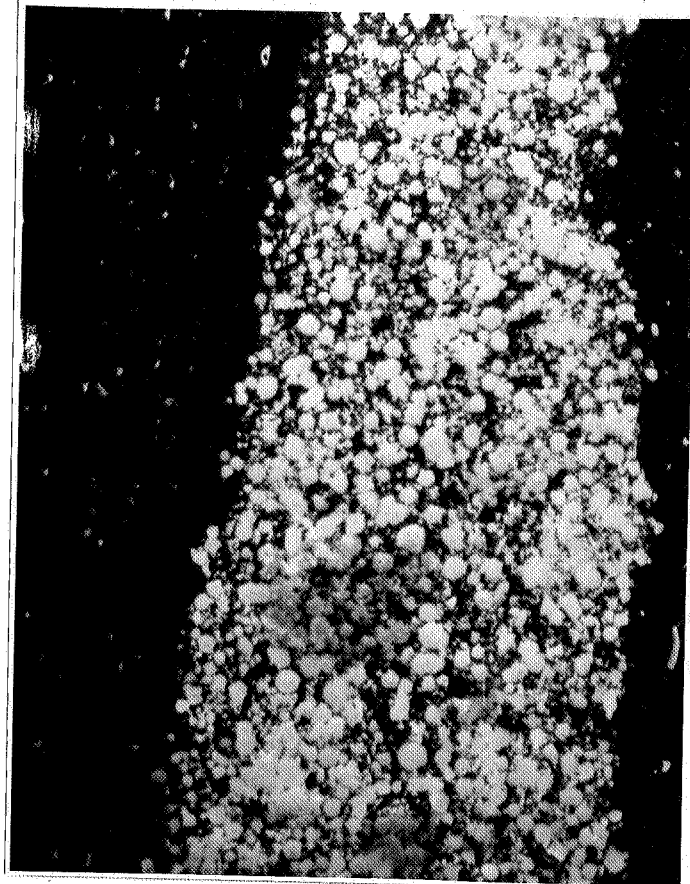
Fine laminated shale.

T.S. Some silty bands (approx. 1 mm wide) with abundant angular to subangular quartz grains (approx. 0.015 mm). Fine illite flakes and kaolinite occur. Also scattered angular sulphide grains (approx. 0.015 mm to 0.04 mm) are seen.

In the shale bands, sulphides are locally abundant as coarse grains (0.03 mm to 0.08 mm) and are concentrated in bands. Erratics of coarse angular > rounded sulphides (approx. 0.03 mm) and fine opaque dust occur scattered throughout abundantly and slightly larger scattered angular > rounded quartz grains occur in a fine kaolinite > illite matrix.

Fine parallel to bedding brownish limonitic dusty bands occur throughout the shale.

000787



0.2 mm

Figure 13 DH Gully 15, 21.80 m (A9740)

A band of framboidal pyrite grains, most of which have been replaced almost entirely by chalcocite (white) or goethite (grey). Possible grains of greigite (bright white), also occur.

Plain light

Magnification X 80

A lenticle (2.5 mm wide) of sulphide with minor quartz inclusions occurs which has disturbed shale lenticles.

P.S. The lenticle of sulphide described above contains recrystallized pyrite framboids at centre. The exterior portions of the lenticle have been grossly replaced by massive bornite and in places by chalcocite.

Scattered pyrite and fewer bornite grains occur 0.003 mm to 0.050 mm, median 0.008 mm. The shapes of many of the grains are quite angular and are somewhat similar in size and shape with adjoining quartz grains.

Only the lenticular copper sulphides would be recoverable here but complete separation from pyrite would be difficult.

Drill Hole - Gully 15

21.80 m (A 9740). Laminated shale. (Figure 13).

No specimen chip is available.

T.S. Similar to specimen at 24.40 m (A 9739).

Mainly shaley bands which are packed with very fine angular quartz (0.005 mm to 0.030 mm) mainly in finer ranges. Some erratic rounded > angular quartz (0.03 mm to 0.06 mm) and similar subrounded sulphides of similar sizes or somewhat smaller in a groundmass of illite and kaolinite. These bands contain fine parallel brown laminae of limonite.

The shale bands grade into silty bands of angular quartz (0.01 mm to 0.015 mm), fine cherts, a little rounded zircon and larger quartz grains mainly

angular (approx. 0.05 mm), also some narrow slivers of quartz. The groundmass is of illite and kaolinite.

Sulphides (0.01 mm to 0.05 mm) are round to subangular and occur to 7 percent in coarser quartzose bands. They occur less commonly in shaley bands but here diagenetic sulphide addition has joined several grains into lamellae distributed along bedding.

A pocket of originally separate grains of sulphide, (1 mm x 2.8 mm) shows diagenetic sulphide precipitation with fusion of grains leaving relict detrital clay interstices.

P.S. A lenticle of chalcocite pseudomorphing pyrite framboids occurs (some relict pyrite is seen) and has some association with greigite (?). Individual grains, which are somewhat rounded, vary to 0.05 mm.

Scattered greigite (?) grains (some contain a little replacive bornite) are seen of similar size to pyrite which occurs in this series.

Some chalcocite grains replacing bornite are seen, grains (approx. 0.015 mm) with a variation of 0.001 mm to 0.10 mm.

Abundant sulphide recovery is possible here.

Drill Hole - Gully 16

24.50 m (A 9741). Laminated shale.

Laminated shale with silty bands occur.

T.S. Sulphide is more abundant here than in previous sections examined. Grains are quite angular to 0.08 mm; or longer and lenticular, to 0.02 mm wide in quartzose bands. Angular to rounded quartz

grains occur of similar size to sulphides in a groundmass of fine illite > kaolinite.

Shaley bands contain finer angular quartz (approx. 0.005 mm) as well as larger subrounded erratics of sulphides and quartz as seen in the silty quartz bands. Fine brown parallel laminae of limonite occur.

Abundant subrounded sulphides (approx. 0.02 mm) are fairly equigranular and occur more commonly in the more quartzose bands.

P.S. Very similar to the 'dispersed' sulphide areas of D.H. Gully 15, 21.80 m (A 9740). Subrounded chalcocite > bornite grains occur (0.001 mm to 0.05 mm, median 0.008 mm) but in the more silty bands. Many grains show internal or peripheral pyrite sphere inclusions showing that much of the copper sulphide is the result of replacement of pyrite framboids. Large numbers of fine framboids of pyrite (approx. 0.001 mm) occur between bands of coarser chalcocite and bornite.

Abundant copper sulphide is present but much of it is too fine for good recovery. Three lenticles of coarse pyrite framboids occur and are replaced by chalcocite where median grain size is 0.08 mm, maximum size is 0.4 mm.

3.2 MG 14 Area

Drill Hole - MG 62

41.20 m (A 9705). Dololutite.

Collenia dolomite as parallel growths alternate with dolomite breccia zones containing bornite.



0.3 mm

Figure 14 DH MG 62, 42.0 m (A9706)

Chalcopyrite (bright yellow) occurs with dolomite inclusions (dark brown) or is intergrown with carrollite (very pale yellow here). Bornite (orange) also occurs as inclusions in, or is attached to massive carrollite.

Plain light

Magnification X 73

T.S. Parallel Collenia growths contain sulphide specks from 0.002 mm to 0.05 mm, median 0.007 mm in dolomite, recrystallized clear patches occur between algal forms. In the dolomite breccia occurs much coarser sulphide up to 0.25 mm but with some very fine (approx. 0.002 mm) material.

Sulphide present is fissure or space filling with secondary carbonate, and is also replacive of calcite/dolomite.

Collenia dolomite and breccia; the rock is pure carbonate except for sulphides present.

P.S. Abundant very fine sulphide occurs down to 0.001 mm but size varies to 0.8 mm size. Most sulphide present is bornite with some bornite:sphalerite, bornite:digenite and bornite:chalcocite intergrowths. Many of the finer (approx. 0.02 mm) grains in the matrix (chalcocite with bornite) show relicts of tiny pyrite spheres. Fine sphalerite (approx. 0.05 mm and less) occurs in the matrix. The coarser sulphides are intergrown with coarser carbonate grains. Also sphalerite selvages (0.008 mm to 0.02 mm thick) occur around many of the larger bornite grains.

Sulphides present are syngenetic pyrite; epigenetic bornite, sphalerite and chalcocite.

42.0 m (A 9706). Dololutite. (Figure 14).

Collenia breccia. Chalcopyrite fills pale carbonate spaces.

T.S. An intraformational Collenia breccia of dolomite fragments with calcite/dolomite cement associated with lenticular fill and replacement sulphides. The sulphides vary from < 0.001 mm to over 1 mm, median 0.003 mm. Where the sulphide fills spaces between breccia fragments it has crystallized as

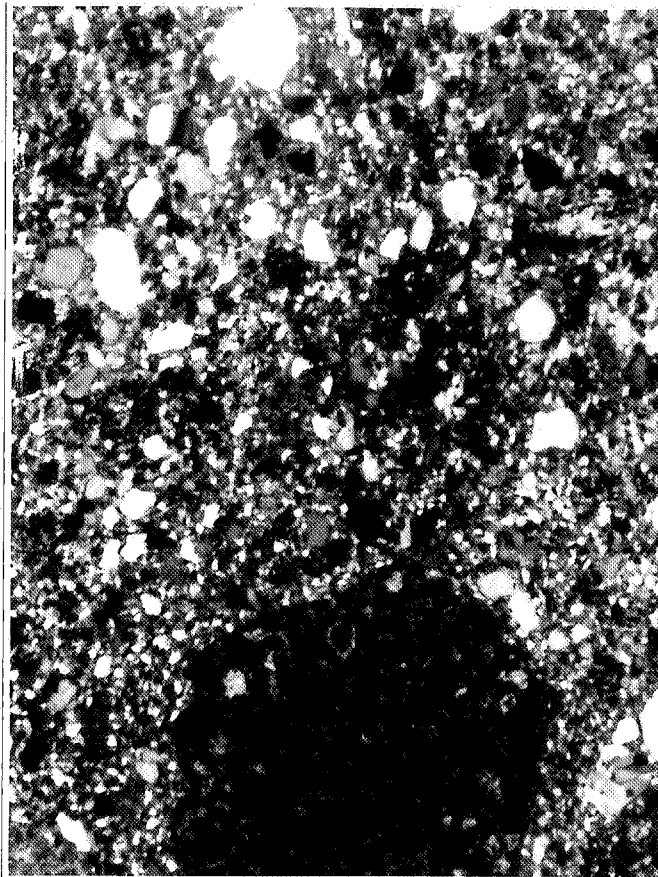


Figure 15 DH MG 63, 39.60 m (A9714), Basal Whyalla Sandstone

A large limonite-stained dolomite fragment of algal forms (bottom of picture). Rounded and subangular grains of quartz (white and grey) and fragments of dolomite (speckled yellow and brown) occur in a fine carbonate (yellow and brown) matrix.

Crossed nicols

Magnification X 14

mutual intergrowths with calcite. A little detrital quartz (0.01 mm to 0.07 mm) occurs in fractures and fissures.

A Collenia dolomite breccia.

P.S. The larger sulphide grains (to 0.80 mm) are mainly chalcopyrite or chalcopyrite:carrollite complex composites. The carrollite commonly contains bornite inclusions (<0.01 mm to 0.04 mm). These sulphides have grown through diagenetic precipitation, are irregular in outline and contain numbers of unreplaced quartz fragments (0.04 mm to 0.06 mm). Some very fine sulphides occur which are mainly bornite, chalcopyrite or composites of these minerals and chalcocite. They vary in size but the median size is around 0.01 mm.

In the finer sulphides may be seen tiny (approx. 0.001 mm) spheres of pyrite. Sphalerite has narrow selvages (approx. 0.003 mm wide) of chalcopyrite. Several small grains of sphalerite (approx. 0.01 mm) as fine grains occur.

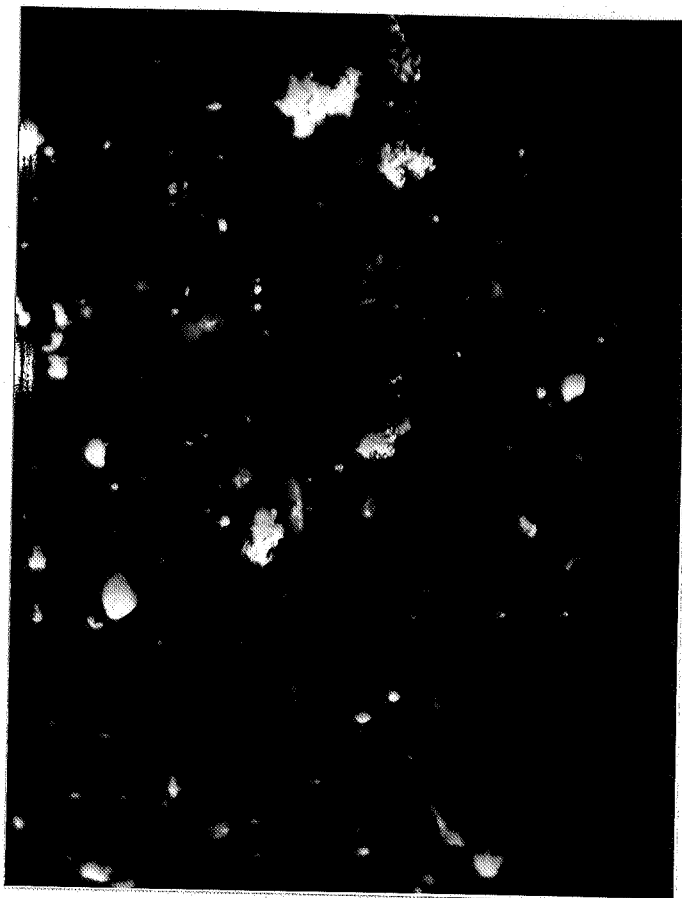
In this specimen occur syngenetic pyrite, epigenetic (diagenetic) copper and zinc sulphides.

Much of the coarse sulphide is recoverable but much of the fine sulphide is locked to pyrite or sphalerite. Abundant carrollite occurs associated with the chalcopyrite as complex locks.

Drill Hole - MG 63

39.60 m (A 9714). Basal Whyalla Sandstone with angular and rounded pebbles. (Figure 15).

Sub-parallel lamellae of carbonate; rounded and subangular quartz pebbles in a fine silty matrix.



0.1 mm

Figure 16 DH MG 63, 39.90 m (A9715)

Scattered angular sulphide grains of bornite (orange) and chalcocite (blue). Some pyrite relicts occur as tiny spheres in the copper sulphide grains.

Plain light

Magnification X 330

T.S. The section contains rounded but flattened limonite stained dolomite of Collenia fragments and oncoliths, ooliths and unstructured dolomite breccia. The pebbly dolomite fragments vary from one to several millimetres in size. Otherwise present are rounded and flat round (also some subangular) quartz rock fragments of irregular quartz mosaics and quartz mineral grains vary from 0.3 mm to 2.5 mm. These are normally more oval than spherical where rounded. The groundmass consists of a second population of subangular quartz grains mainly varying from 0.05 mm to 0.08 mm in a fine carbonate matrix. Some opaque mineral (approx. 0.03 mm) occurs of subrounded grains to about 1 percent with the silty matrix.

Refer to MG 21, 48.60 m (Reference 3).

P.S. Traces of chalcopyrite as clusters of tiny grains (0.004 mm to 0.03 mm) occur along a channelway in a large quartz fragment and also cementing a little silt size quartz matrix. Elsewhere, only goethite is seen (possibly oxidized pyrite) in the matrix, as small grains.

39.90 m (A 9715). Black shale. (Figure 16).

A very fine grained grey shale rock with abundant sulphides.

T.S. Rounded to angular chips of quartz (0.002 mm to 0.05 mm) unoriented in a fine groundmass of kaolinite and illite the most abundant minerals present and a little irregular fine carbonate (0.002 mm to 0.01 mm). Euhedral to irregular sulphide forms (0.002 mm to 0.05 mm, with a median grain size of 0.005 mm) occur to 2 to 3 percent. Scattered flakes of illite (apart from the very fine mineral) occur to 0.02 mm x 0.005 mm.

One 'large' lenticle of sulphide intergrown with carbonate occurs (2.5 mm x 1 mm). The sulphides

indicate a very rough banding where they occur in slightly greater abundance. A faint banded dark clay.

P.S. Fine, scattered, angular sulphides (0.003 mm to 0.015 mm, median 0.005 mm) of bornite > chalcocite. Occasional replacements of pyrite framboids are seen by bornite and chalcocite but these are rare. Most sulphide grains are quite pure.

Probably sulphides are too fine for good flotation. Recovery rate would be unacceptable.

Drill Hole - MG 64

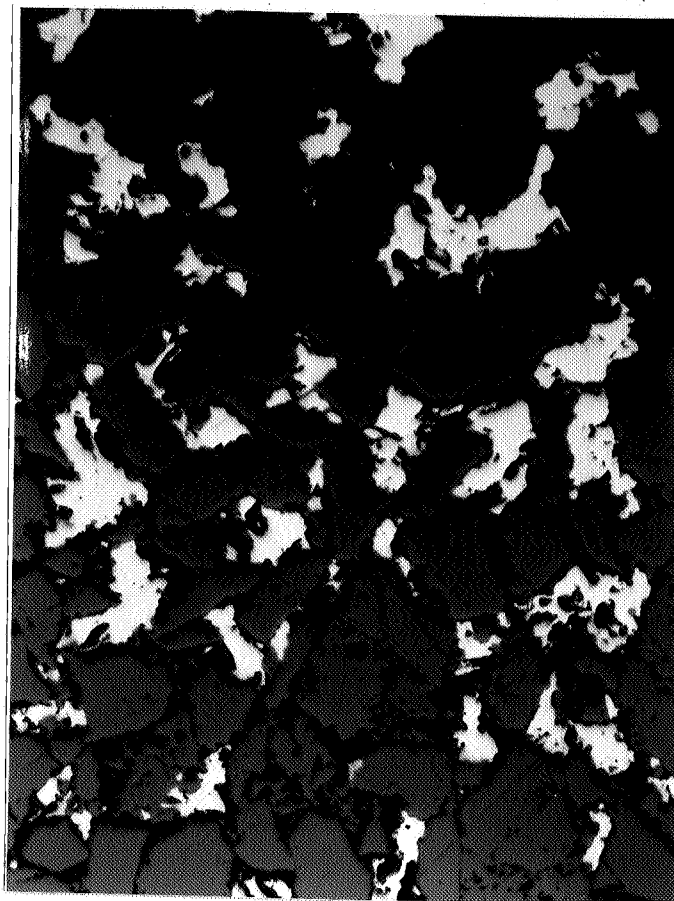
33.30 m (A 9707). Grey dolomite.

Brecciated dolomite of Collenia growths. Most specks of sulphides occur in matrix. Some specks are in Collenia dolomite fragments.

T.S. Intraformational brecciation of dolomite Collenia growths has occurred. A fine carbonate matrix (mainly calcilutite) contains sulphides as irregular grains (approx. 0.03 mm) and ribbons to 0.2 mm x 0.05 mm replacing calcite grains. In the Collenia lamellae (1 mm to 2 mm wide) sulphides occur as ribbons as above but also as very fine grains <0.02 mm, scattered around dolomite mass. No quartz is seen. Sulphides occur from 2 to 3 percent.

P.S. Scattered, irregular sulphides occur including bornite and carrollite in the coarser sizes; bornite, carrollite, chalcocite and a little sphalerite with some pyrite in the finer sizes. Size varies 0.005 mm to 0.27 mm with a median size of 0.035 mm.

Many grains, even the finer ones are complex composites of copper sulphides with carrollite > sphalerite. Sphalerite selvages (0.02 mm wide) occurring around bornite is an example.



0.3 mm

Figure 17 DH MG 64, 38.00 m (A9710)

Chalcopyrite (yellow) occurs in spaces
between angular quartz grains (yellowish-
brown).

Plain light

Magnification X 80

No worthwhile cobalt or zinc free concentrate could be prepared from this ore horizon. Abundant carrollite occurs locked mainly with bornite.

Sulphides present are epigenetic except for relict syngenetic pyrite.

33.35 m (A 9708). Black shale.

Very fine dark grey shale which effervesces briskly with 1:1 HCl, with no odour. Calcite is present.

T.S. The rock consists largely of fine irregular calcite grains (<0.001 mm to 0.023 mm), less interstitial kaolinite flecked with minute (<0.001 mm) black (carbonaceous ?) specks and 5 to 10 percent of sub-angular quartz (0.003 mm to 0.05 mm, median 0.02 mm). Sulphides occur as subrounded and subangular, with some sharply angular grains (<0.002 mm to 0.1 mm, median 0.007 mm) probably diagenetically altered syngenetic, though some grains may be detrital as indicated by the subrounded shapes ?

P.S. Scattered fine bornite > chalcocite grains occur which replace centres of framboidal pyrite. The grain sizes vary from <0.001 mm to 0.030 mm, median 0.015 mm. Occasional bornite:chalcopyrite intergrowths free from pyrite occur, and some mainly free chalcocite where only included scattered spheres of pyrite are left as relicts. Replacement of syngenetic pyrite by copper sulphides is seen.

38.00 m (A 9710). Black sand cement, dolomite fragments. (Figure 17).

Shows brecciation of Collenia dolomite lamellae.

T.S. The rock consists of lamellae of dolomite breccia in a matrix of fine dolomite, subangular to angular quartz (few rounded grains, range from 0.05 mm to

0.7 mm, median 0.2 mm) in a dolomite matrix. Many quartz grains are encapsuled by an even zone (approx. 0.03 mm thick) of dolomite. There are oolites of dolomite (approx. 0.2 mm) sometimes but not always with an obvious quartz nucleus. Chert grains and quartz rock fragments (irregular mosaics) are present but much less so than quartz mineral grains. Lenticles of quartzite occur with just a smear or no carbonate cement between quartz grains.

Sulphides occur sporadically and mainly as irregular grains around 0.003 mm, though some larger sizes are seen in dolomite. Otherwise sulphides occur as cement between quartz grains in 'quartzite' lamellae (<0.01 mm to 0.3 mm x 0.2 mm).

Dolomite breccia fragments occur with fine disseminated syngenetic sulphides in dolomite or as epigenetic replacement sulphides in 'quartzite'. There are some similarities with A 9962 (80115) MG 14 - 31.25 m in thin section, (Reference 2).

P.S. Chalcopyrite occurs patchily as medium to coarse interstitial grains (0.09 mm to 0.3 mm) between coarse dolomite fragments. Many of the coarser chalcopyrite grains have narrow selvages of sphalerite (0.02 mm to 0.04 mm wide) between the copper sulphides and adjoining dolomite fabric.

One large bornite:chalcopyrite grain is seen (approx. 0.08 mm). In the fine dolomite lamellae are inclusions of very fine sulphides, a little bornite and chalcopyrite is seen replacing pyrite framboids but mainly recrystallized cubes of pyrite (approx. 0.005 mm) are seen.

Trails of pyrite framboids parallel to 'banding' are seen.

Recovery of the coarse chalcopyrite will also bring abundant sphalerite which occurs as selvages to copper minerals.

Drill Hole - MG 65

30.25 m (A 9709). Brown sand/black shale.

A creamy carbonate rock specked with sand grains.

T.S. A sandy dolarenite of irregular to oval oncolith (algal) fragments which average around 0.14 mm in a calcite cement. Oolites of dolomite occur but not commonly with nuclei of quartz. Quartz grains present vary 0.08 mm to 0.35 mm, median 0.1 mm and though in the dolarenite layers many rounded or subrounded forms occur some grains are angular to sharply angular, particularly the larger grains.

A clay-carbonate band (2 mm wide) occurs of fine kaolinite and carbonate with many quartz grains around 0.015 mm. Algal fragments and larger quartz grains occur as in the main dolarenite but there calcite cement is present in lieu of kaolinite and fine quartz.

A sandy shallow water deposit from slightly beyond the Collenia reef talus.

Sulphides are seen (up to 0.2 mm x 0.2 mm) replacing the carbonate. Some fine sulphide (approx. 0.015 mm) occurs in dolomite fragments and matrix.

P.S. Consists of scattered large (0.09 mm to 0.36 mm) complex sulphide grains of bornite with sphalerite selvages; bornite with carrollite selvages; bornite: sphalerite complex mixtures; bornite with many fine inclusions of carbonate; and bornite:chalcopyrite composites (0.02 mm to 0.07 mm). Very fine (0.005 mm to 0.02 mm) bornite and chalcocite grains in the

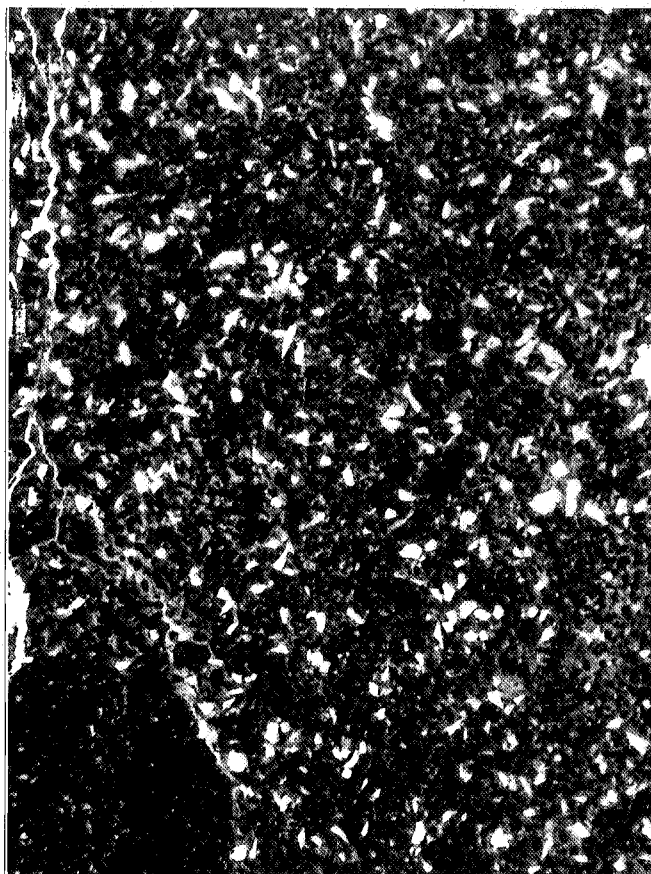


Figure 18 DH MG 65, 30.60 m (A9711)

A dolomitic Collenia (algal) fragment (left, bottom corner of photograph). Sharp angular quartz fragments (white) are intermixed with dolomite (yellowish-green and brown). Carbonate (white) veins traverse matrix of rock. Fine sulphides also occur (black).

Plain polarised light Magnification X 16

groundmass are very abundant and many of them contain fine pyrite spheres (approx. 0.001 mm to 0.002 mm).

A difficult ore for beneficiation, but much of the middle size bornite and chalcocite would be recovered free from sphalerite, carrollite, etc.

Mainly epigenetic cavity-filling sulphides.

30.60 m (A 9711). Brown sand with brown dolomite. (Figure 18).

Similar to MG 65, 30.25 m (A 9709), but larger lenticles of carbonate are included with dolarenite mass.

T.S. A fragment (several millimetres wide) of Collenia reef shows algal structures. Calcite occupies parallel lateral space within the reef structure and also a little angular quartz (approx. 0.05 mm).

Beyond the reef fragment occurs dolarenite of slightly finer grain size. Dolomite fragments vary to 0.15 mm, median 0.06 mm with calcite matrix. Quartz grains 0.01 mm to 0.3 mm, median 0.08 mm are commonly sharply angular and similar to sharks teeth, or are spade shaped and remind one of fine tillitic forms. Some rounded quartz grains (approx. 0.08 mm) also occur. Sulphides (0.02 mm to 0.25 mm) are highly irregular and replace carbonate (fragments of Collenia and matrix). Some sulphide grains have a fragmented appearance.

P.S. Coarse sulphides (up to 0.25 mm but some lenticles to 1 mm long) occur associated with some coarse dolomite layers. The sulphides vary from 0.02 mm and are mainly about 0.1 mm. They are bornite or bornite complex locked with carrollite or chalcocite. Also some bornite occurs and bornite with chalcocite locked to coarse carbonates, embedded in a mosaic of carbonate grains. Some of the larger bornite

grains have a selvage of sphalerite (varying 0.005 mm to 0.05 mm thick).

Very fine sulphides (0.005 mm to 0.01mm) are scattered around the rock matrix and are chalcocite, bornite or composites of the minerals. Pyrite framboids also occur. Many of the chalcocite or chalcocite: bornite grains are perfectly spherical (replacements of perfect pyrite spherical framboids or detrital grains ?). Few of the spherical copper sulphide grains contain pyrite spheres, mainly the more angular ones which show an obvious growth pattern.

Coarser sulphide lenticles and few fine sulphides occur in Collenia bands.

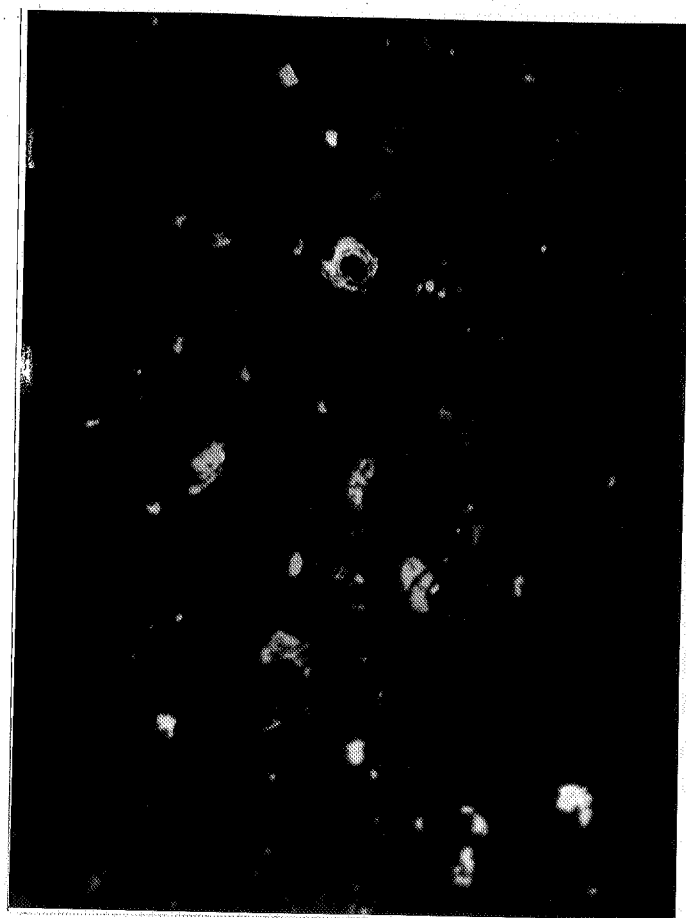
Some residual goethite grains are angular.

36.0 m (A 9712). Black sand with dolomite fragments.

Similar to 30.60 m (A 9711) but with parallel Collenia reef fragments alternating with dolarenite containing quartz.

T.S. In the lamellae of reef, parallel algal growths occur with calcite alternating with reef forms and occasionally quartz fragments (0.005 mm to 0.1 mm) as subangular to rounded detrital grains. Sulphides occur as irregular masses of fill and replacement forms (up to 0.05 mm x 0.15 mm) but many specks around 0.005 mm occur.

In the dolarenite the Collenia oncoliths and oololiths vary up to 0.3 mm, median 0.08 mm. Many of the oololiths are concavo-convex/ovoid shapes with a limonitized carbonate capsule (approx. 0.015 mm wide). Quartz and chert grains are of similar median size but with larger maxima to 0.45 mm. Some are quartz rock fragments with an irregular



0.2 mm

Figure 19 MG 65, 36.90 m (A9713)

Replacement of pyrite framboids (yellow) by bornite (orange-pink) is seen. Other unreplaced pyrite and some very fine bornite grains occur.

Plain light

Magnification X 150

quartz mosaic, many are quartz mineral grains. Most are rounded to subrounded but there are large angular grains. Some quartz grains show marginal replacement by carbonate. Sulphides are mostly irregular replacive forms though some show detrital shapes. They mostly replace carbonate and are mainly around 0.02 mm in size.

The specimen is part of reef talus adjoining the reef.

P.S. Coarse chalcopryite (up to 0.4 mm) occurs in the interstices of dolarenite bands. Traces of sphalerite occur at edges of chalcopryite crystals. Smaller bornite grains (approx. 0.05 mm) also occur.

In the finer interstitial Collenia bands, many pyrite framboids (0.02 mm to 0.09 mm) occur and some of these show some replacement by bornite or less so by chalcocite.

Some of the finer dolomite bands contain small very round bornite:chalcopryite composites around 0.03 mm, some as small as 0.005 mm which are usually less rounded and more subangular.

Very little coarse copper sulphide is present but this is mainly recoverable.

36.90 m (A 9713). Silt, shale band, as in MG 64 (92933). (Figure 19).

Silty/sandy and carbonate bands.

T.S. The silty/sandy layers consist of unsorted angular subangular and rounded quartz grains varying between 0.01 mm and 0.1 mm, in a carbonate and illite matrix, with fragments of carbonate to 0.5 mm x 0.5 mm and smaller lenticles of the mineral. Much of the sulphide present (2 to 3 percent) is irregular (0.05 mm

to 0.1 mm) or lenticular (1 mm x 0.05 mm) but finer grains (approx. 0.02 mm) occur and many of these are rounded (possibly detrital ?). The fine carbonate parallel lenticles appear to have Collenia forms intact with moderate specks of sulphide (<0.01 mm to 0.03 mm) and are rare subangular to rounded detrital quartz grains (approx. 0.07 mm).

Finer grained than MG 64, 38.00 m (A 9710/92933), but otherwise similar.

P.S. Abundant framboidal pyrite occurs (0.005 mm to 0.09 mm) shows variable stages of replacement by bornite > chalcocite. The pyrite shows some very round framboids with their interiors replaced by bornite. Occasional subangular bornite or chalcocite grains (0.03 mm to 0.08 mm) are seen with no pyrite relicts.

Sulphides present are mainly pyrite; very little copper sulphide is present.

P.J. CURTIS
(LMB)

4. REFERENCES

1. BINDA, P.L. Detrital bornite grains in the late Precambrian 'B' Greywacke of Mufulira, Zambia. Mineralium Deposita, 10, No. 2, 1975, pp 101-108.
2. Memorandum from P.J. Curtis, to Dr. K.J. Maiden dated 8th January, 1974. Entitled 'Samples from Drill Hole MG 14D containing Copper Mineralisation in Fine Dolomite'.
3. Appendix to 'MG Area Geology'. Pacminex Pty. Limited Report No. PMR 15/75.

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APPENDIX I

MEMO FROM J.M. HEAPE TO P.J. CURTIS 'SAMPLES
FROM GULLY AND MG 14 AREA', DATED 12TH JUNE, 1975.

PACMINEX PTY. LIMITED

000810

MEMORANDUM TO: MR. P.J. CURTIS

OUR REF: JH/LK

FROM: J. HEAPE

DATE: 12/6/75
(typed)SAMPLES FROM GULLY AND MG 14

Sample numbers 92901-92926 from Gully Drilling :-

		<u>HOLE NO.</u>	<u>DEPTH (M)</u>	<u>SAMPLE DESCRIPTION</u>
A 9718	92901	Gully 1	27.80	Laminated shale
A 9717	02	1	27.30	Laminated shale
A 9716	03	1	23.15	Shale/dolomite bands
A 9719	04	2	20.30	Dolomite lens in shale with bornite
A 9720	05	2	20.5	Dolomite/shale bands with bornite
A 9721	06	2	23.20	Brown/green shale
A 9722	07	4	20.55	M.O.C. area chalcocite in P.Q.
A 9723	08	5	21.60	Dolomite/shale bands
A 9724	09	5	22.60	Dolomite shale bands with pyrite, chalcopyrite
A 9725	10	6	30.65-70	Laminated shale and dolomite- arenaceous bands
A 9726	11	6	31.3	Laminated shale, chalcopyrite, bornite, pyrite and desseminated chalcocite
A 9727	12	7	28.80	Laminated shale with chalcocite
A 9728	13	7	29.50	Green clay and quartz pebbles
A 9729	14	8	29.50	Laminated shale
A 9730	15	8	30.80	Purple silt with pebbles

		<u>HOLE NO.</u>	<u>DEPTH (M)</u>	<u>SAMPLE DESCRIPTION</u>
A 9731	92916	Gully 9	34.40	Laminated shale, pyrite, bornite
A 9732	17	11	29.65	Laminated shale, pyrite, bornite
A 9733	18	12	25.0	Laminated shale
A 9734	19	13	27.5-29.0?	Laminated shale
A 9735	20	13	32.8-32.85	Laminated shale with sand lenses
A 9736	21	13	34.5	Laminated shale/silt
A 9737	22	14	23.20	Laminated shale, with mud dyke
A 9738	23	14	24.30	Laminated shale and sulphide
A 9739	24	14	24.40	Laminated shale with pyrite, chalcopyrite, bornite
A 9740	25	15	21.80	Laminated shale with chalcocite, +Ag
A 9741	26	16	24.50	Laminated shale + sulphide

Samples from MG 14 area :-

A 9705	92927	MG 62	41.20	Dololutite (?) bornite, chalcocite
A 9706	28	62	42.0	Dololutite pyrite, chalcopyrite, bornite, chalcocite
A 9714	29	63	39.60	Basal Whyalla sandstone with angular and rounded pebbles
A 9715	30	63	39.90	Black shale and sulphides
A 9707	31	64	33.30	Grey dolomite with chalcocite

	<u>HOLE NO.</u>	<u>DEPTH (M)</u>	<u>SAMPLE DESCRIPTION</u>
A 9708 92932	MG 64	33.35	Black shale
A 9710 33	64	38.00	Black sand cement, dolomite fragments
A 9709 34	65	30.25	Brown sand/black shale
A 9711 35	65	30.60	Brown sand with brown dolomite
A 9712 36	65	36.0	Black sand with dolomite fragments
A 9713 37	65	36.90	Silt, shale band, as in MG 64 (92933)

Samples were selected for maximum sulphide mineral content wherever possible. However sample numbers 92913 and 92915, also 92929 and 92936 were picked mainly for petrographic analysis.

I feel the most benefit would be obtained from these samples if a thin section was prepared from each one, followed by mineragraphic mounts of a selected few. However we can discuss this after you have seen the samples.

J.M. HEAPE
(signed)

c.c. R.W. Gersteling

000813

APPENDIX II

LOCATION MAP OF GULLY DRILL HOLES